

Miscanthus: Abundance-Occupancy and Shared ASVs Report

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About this report

This Quarto report executes two existing analysis scripts:

- 06_alpha_beta_div.R — Community Diversity Analyses
- 07_abund_occupancy.R — Abundance-Occupancy Analyses
- 08_shared_asvs_upset_plots.R — Shared ASVs Analyses

Note

Scripts assume project-root working directory (for sourcing `R/utils/00_setup.R`). Place this `.qmd` at the repository root so relative paths resolve as expected.

If your environment needs it, ensure all packages used by the scripts are installed before rendering.

Render from the repository root:

- CLI: `quarto render mxg_abun_occu_report.qmd --execute-dir .`
- R: `quarto::quarto_render("mxg_abun_occu_report.qmd")`

Section 1 — Abundance-Occupancy Analyses

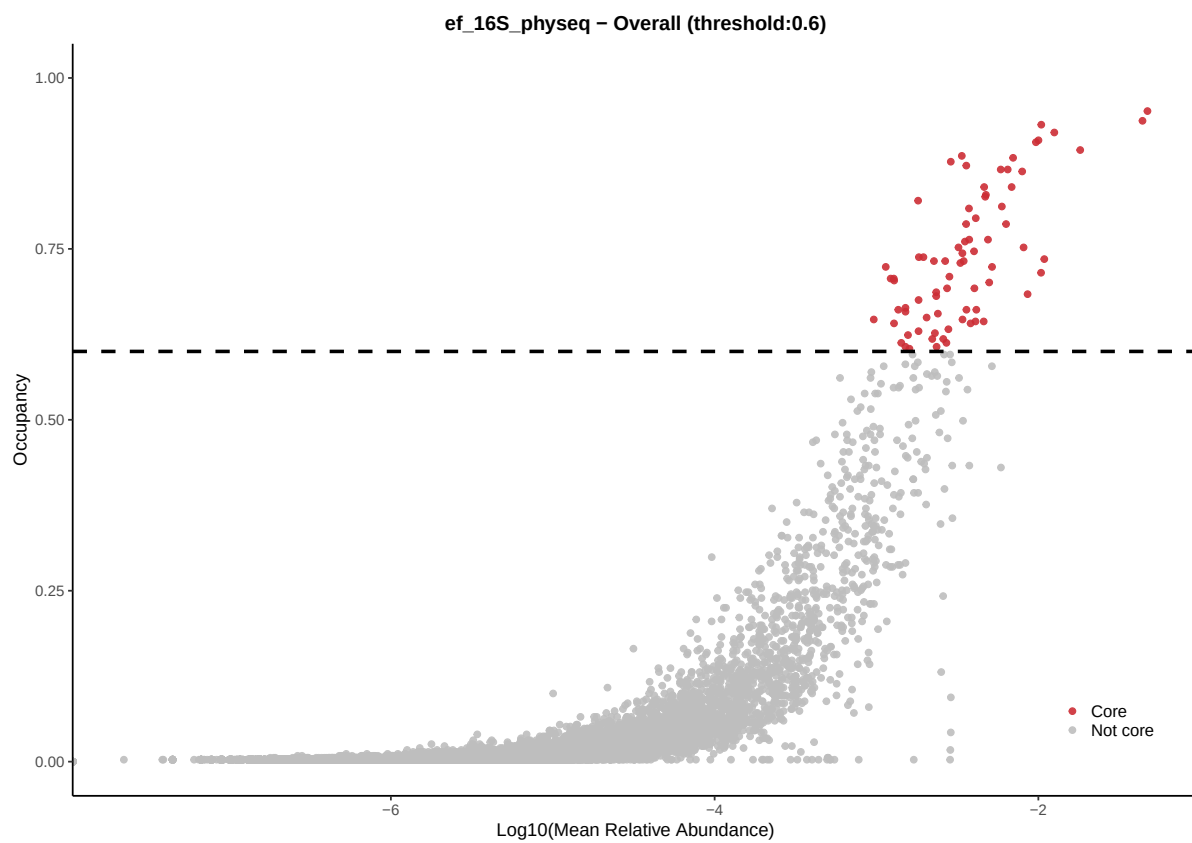
Addresses the following questions:

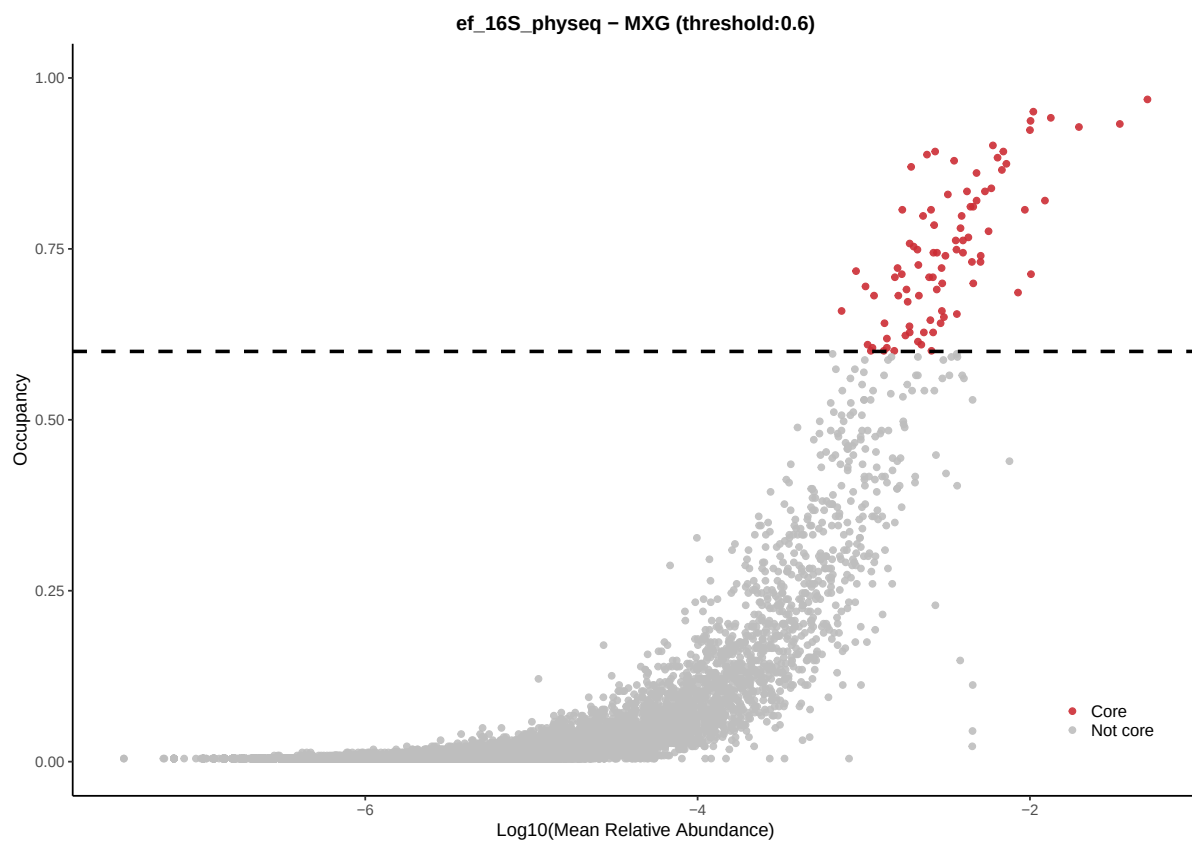
1. What are the abundance-occupancy patterns in *Miscanthus* microbiomes across different projects, regions, and sample types?
2. How do these patterns vary by timing (EF), year (LAMPS 2018), and site (LAMPS 2022)?

This section runs the full workflow in [07_abund_occupancy.R](#), including:

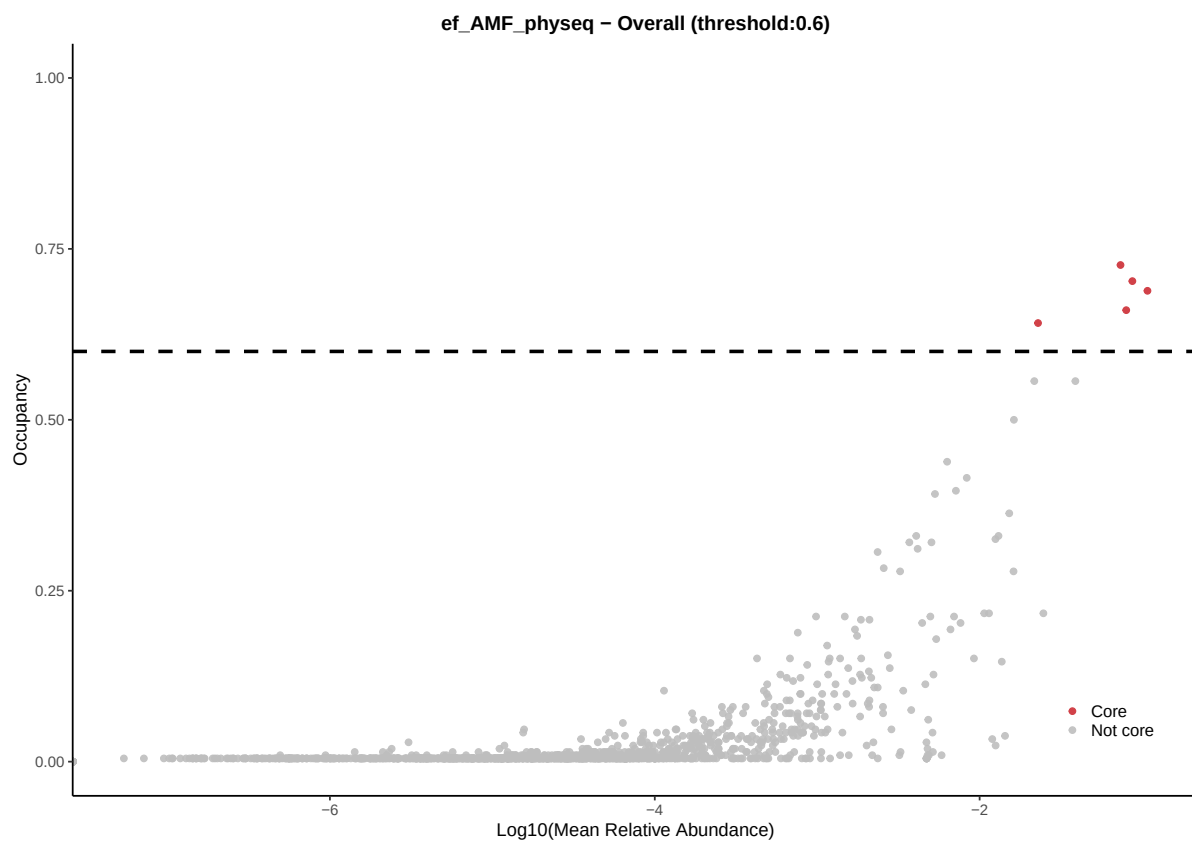
- Abundance-Occupancy plots

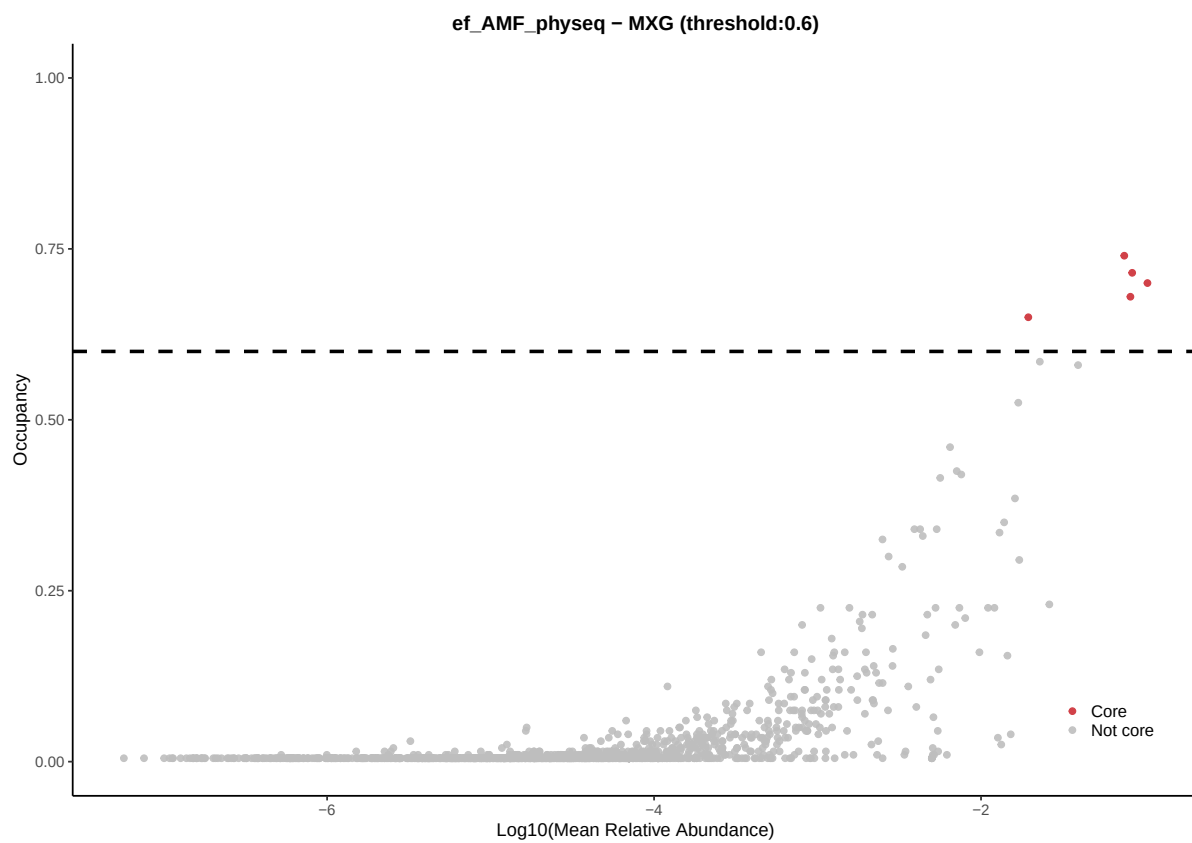
```
source(  
  "R/07_abund_occupancy.R",  
  local = knitr::knit_global(),  
  echo = FALSE,  
  print.eval = TRUE  
)
```



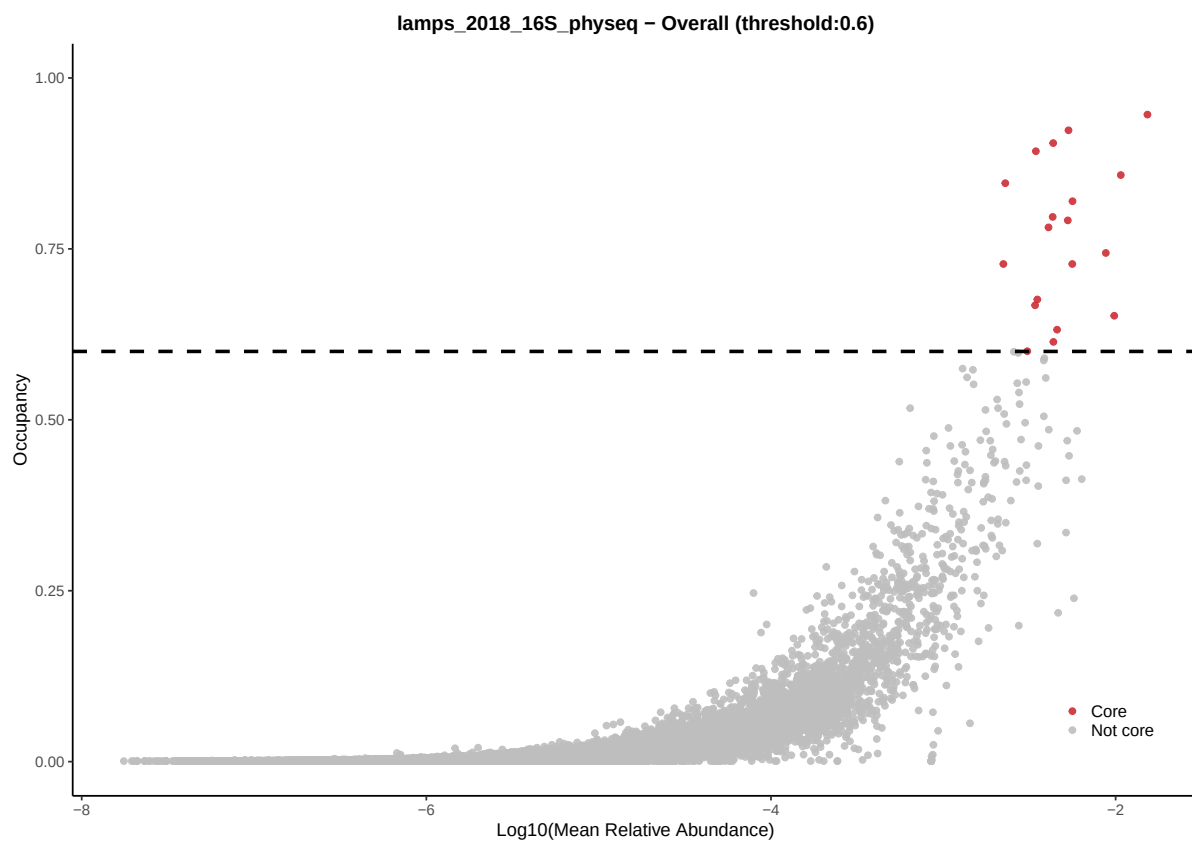


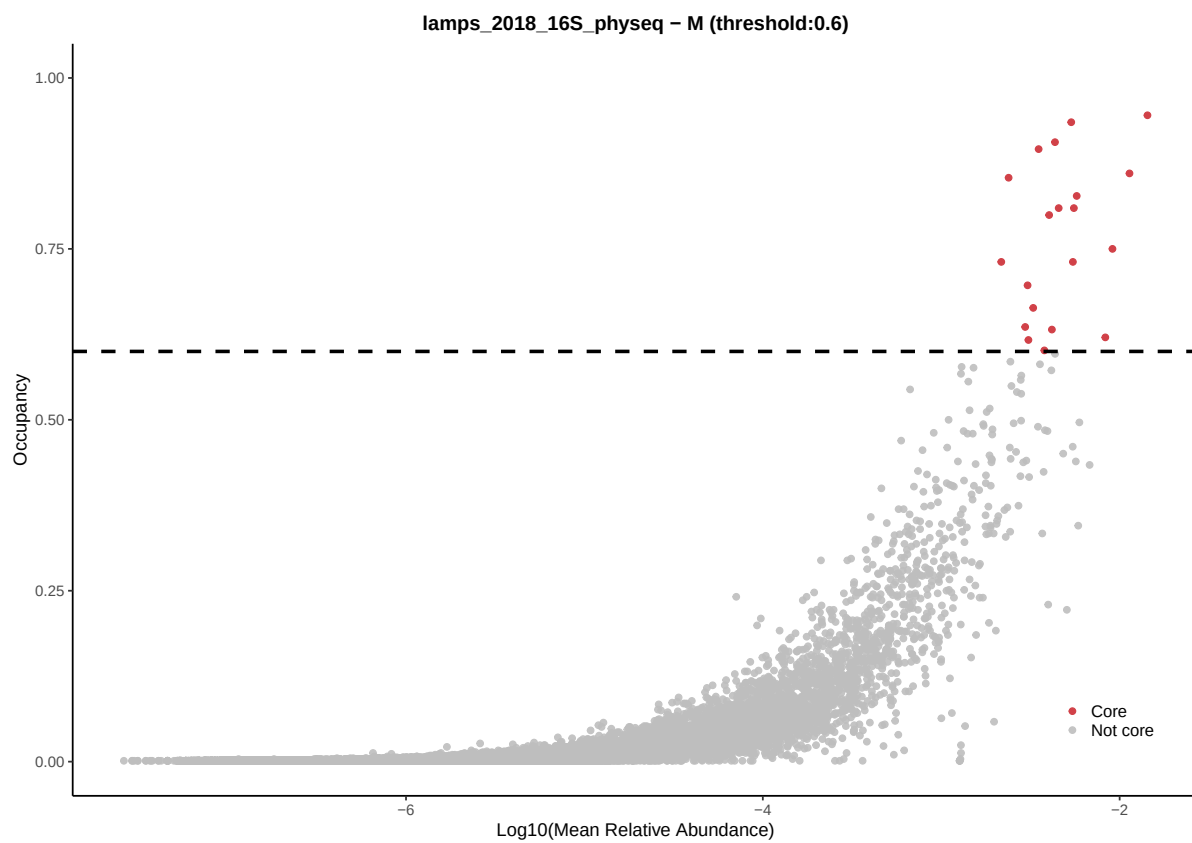
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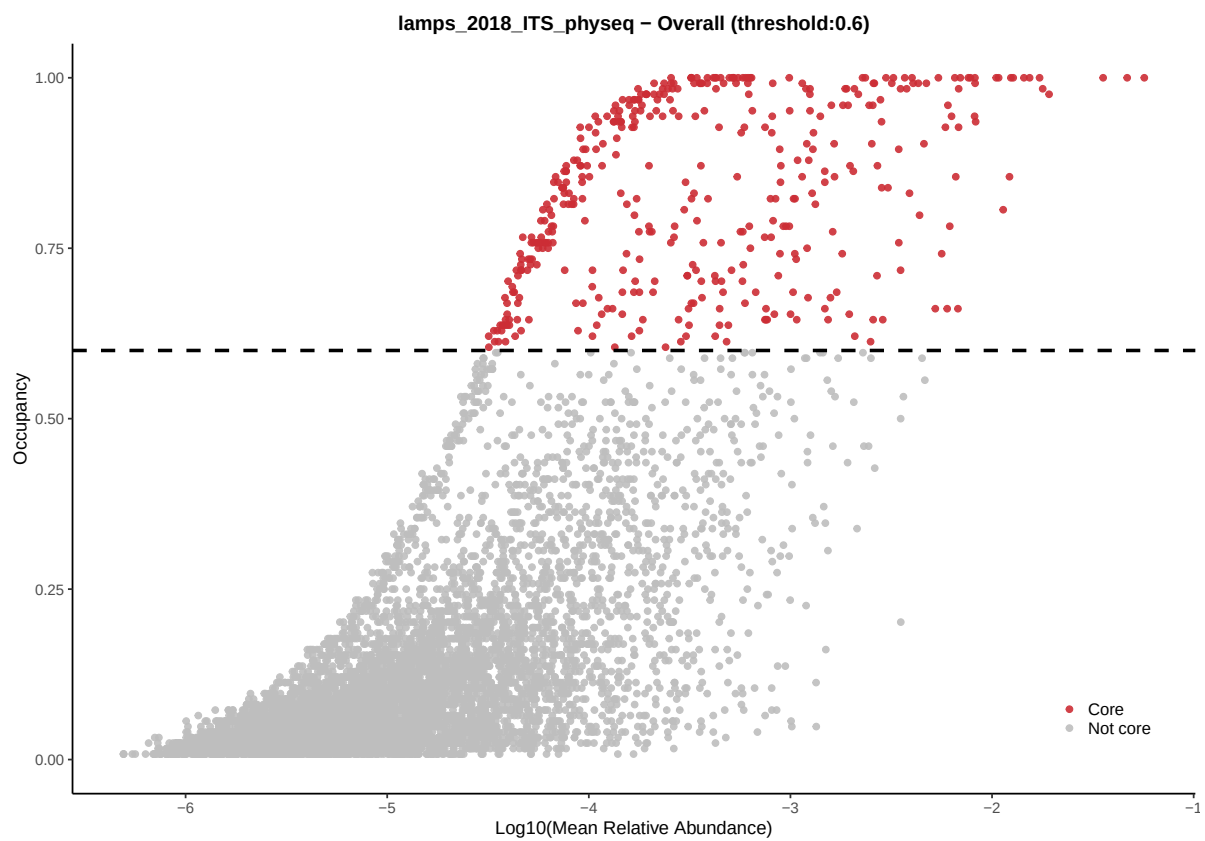


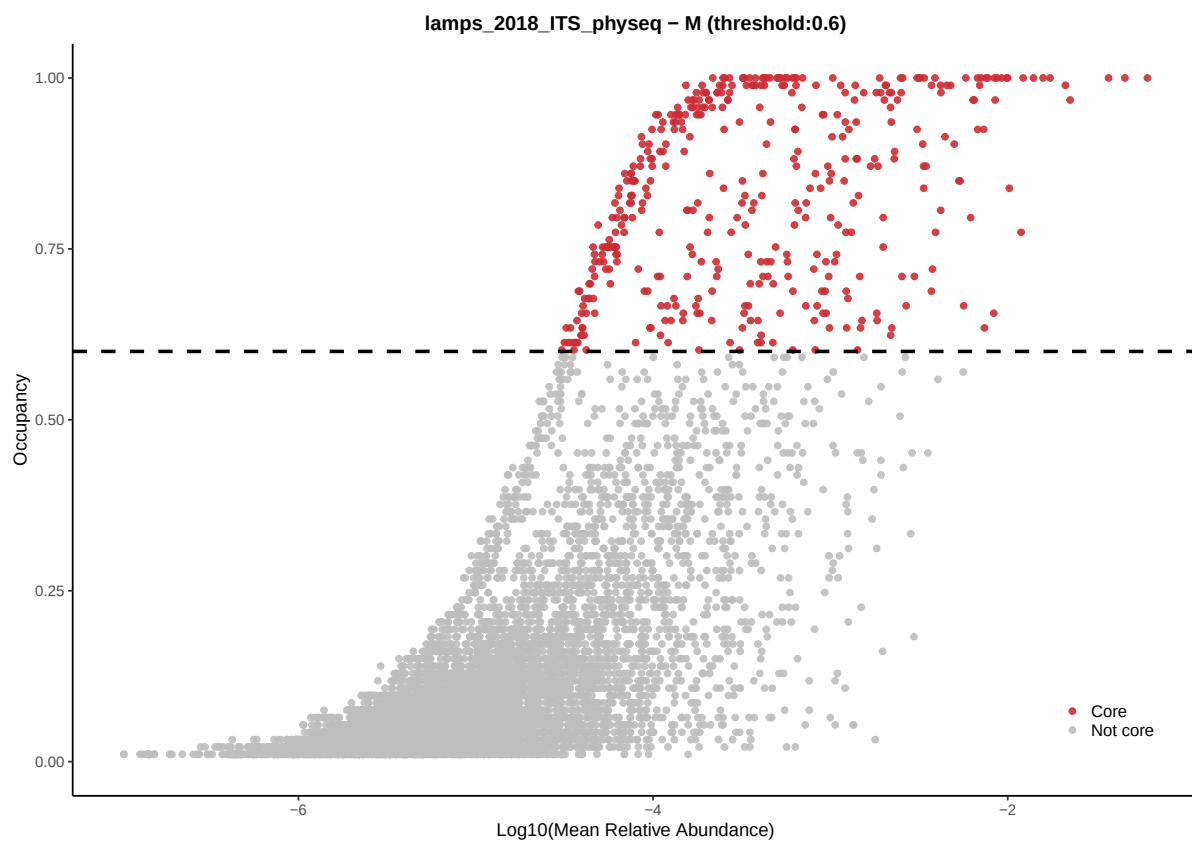
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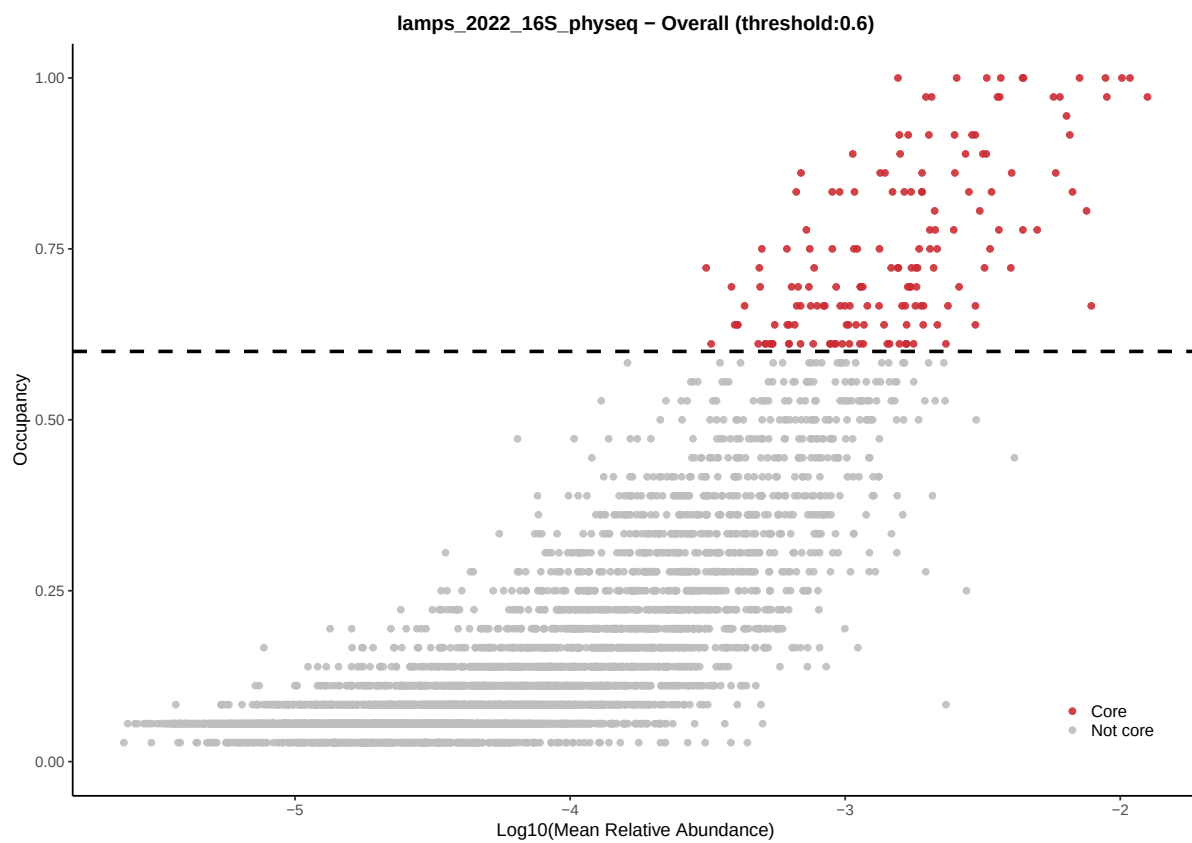


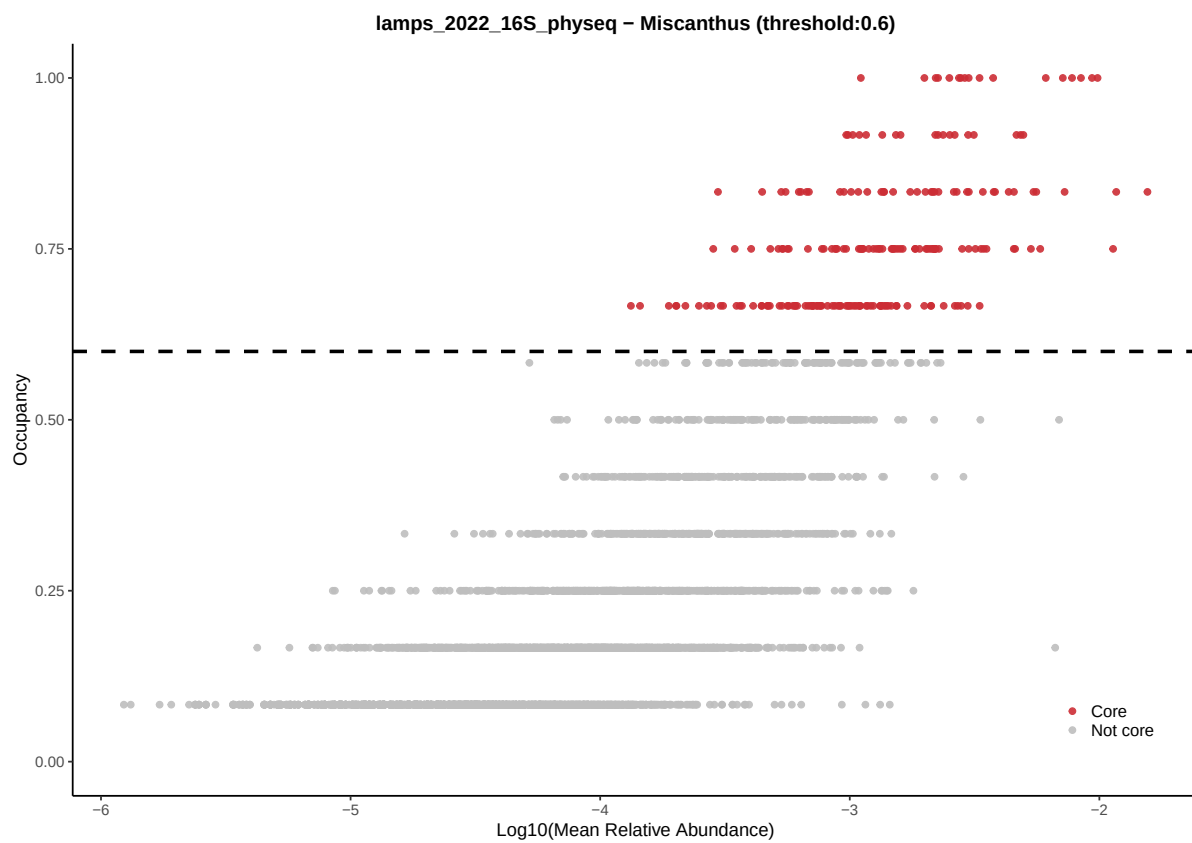
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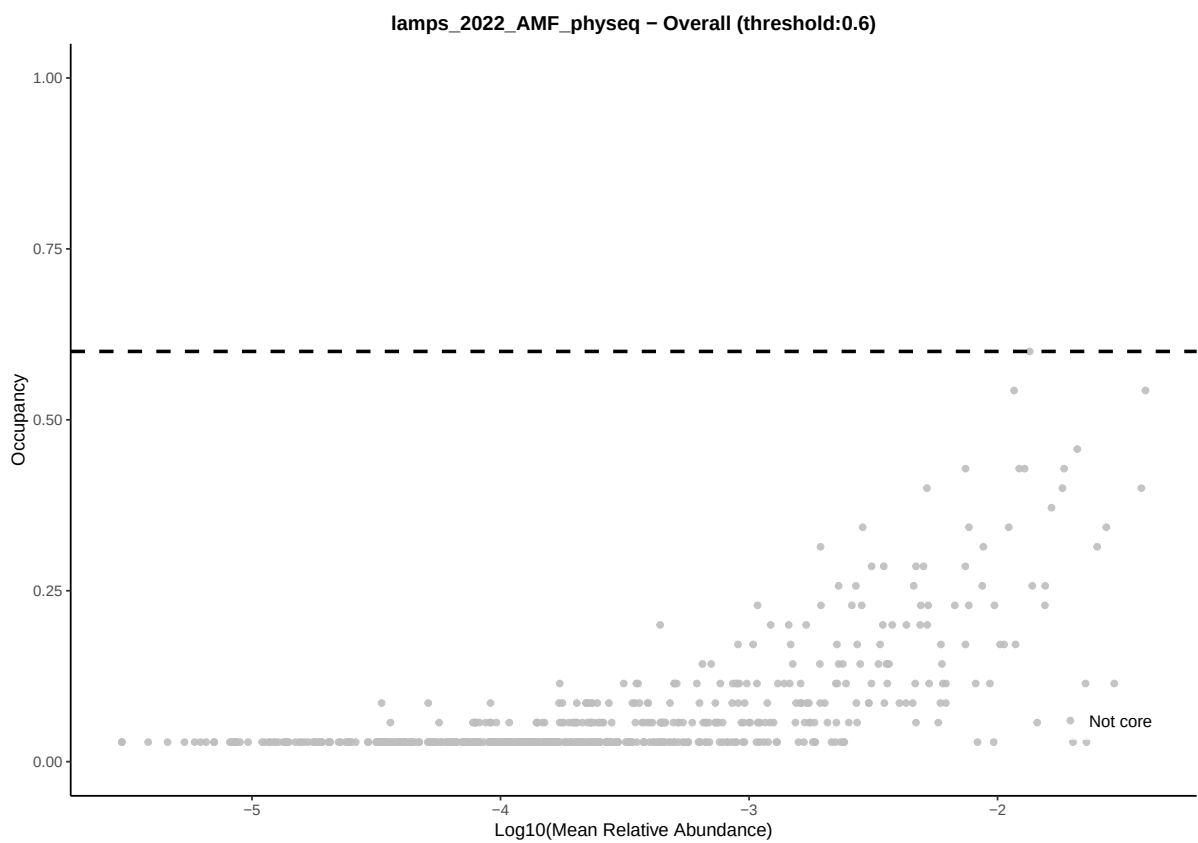


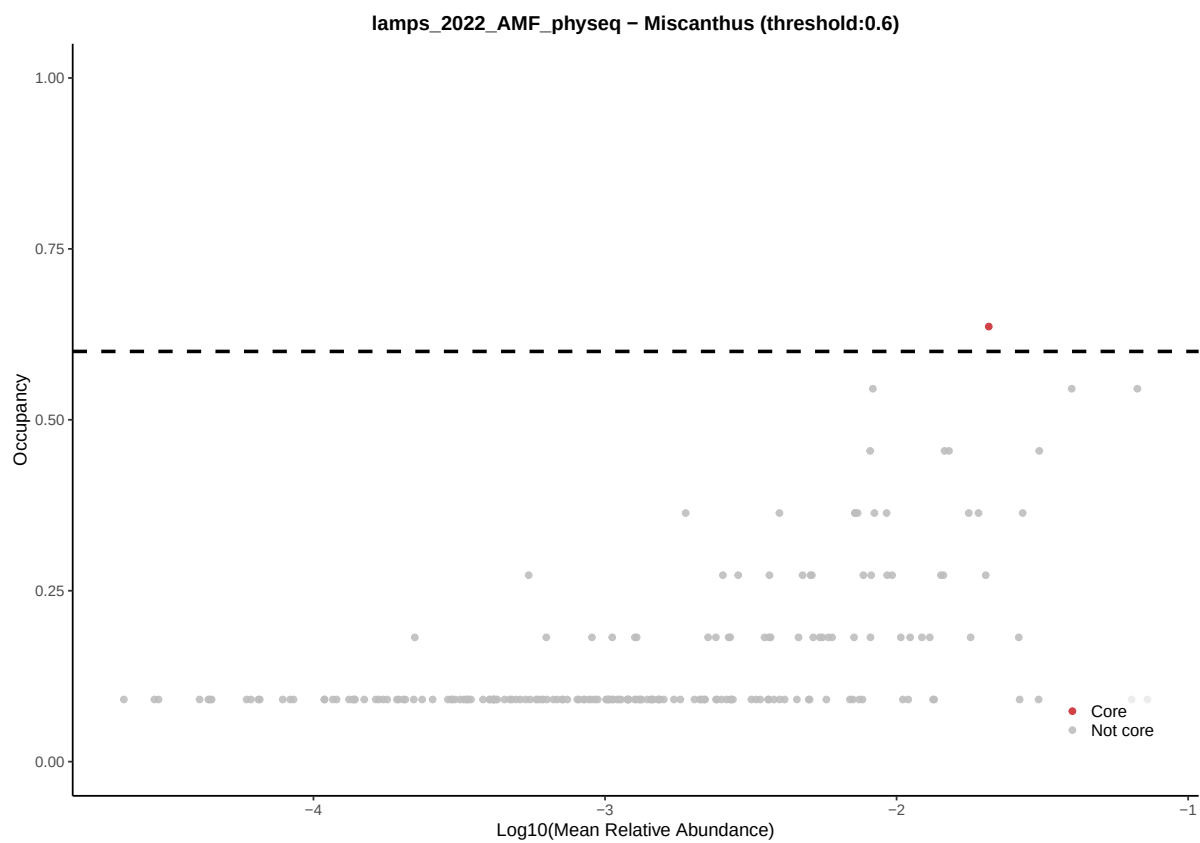
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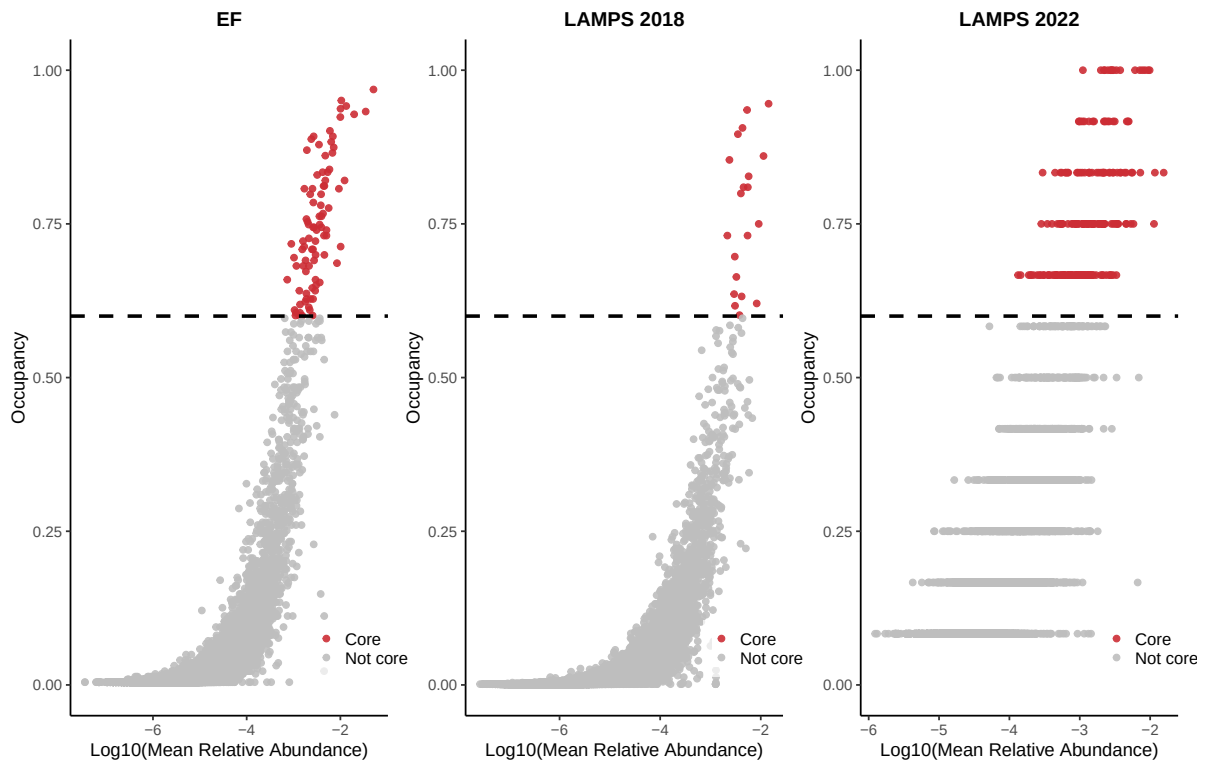




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Abundance–Occupancy Curves: Miscanthus 16S– 3 project(s)

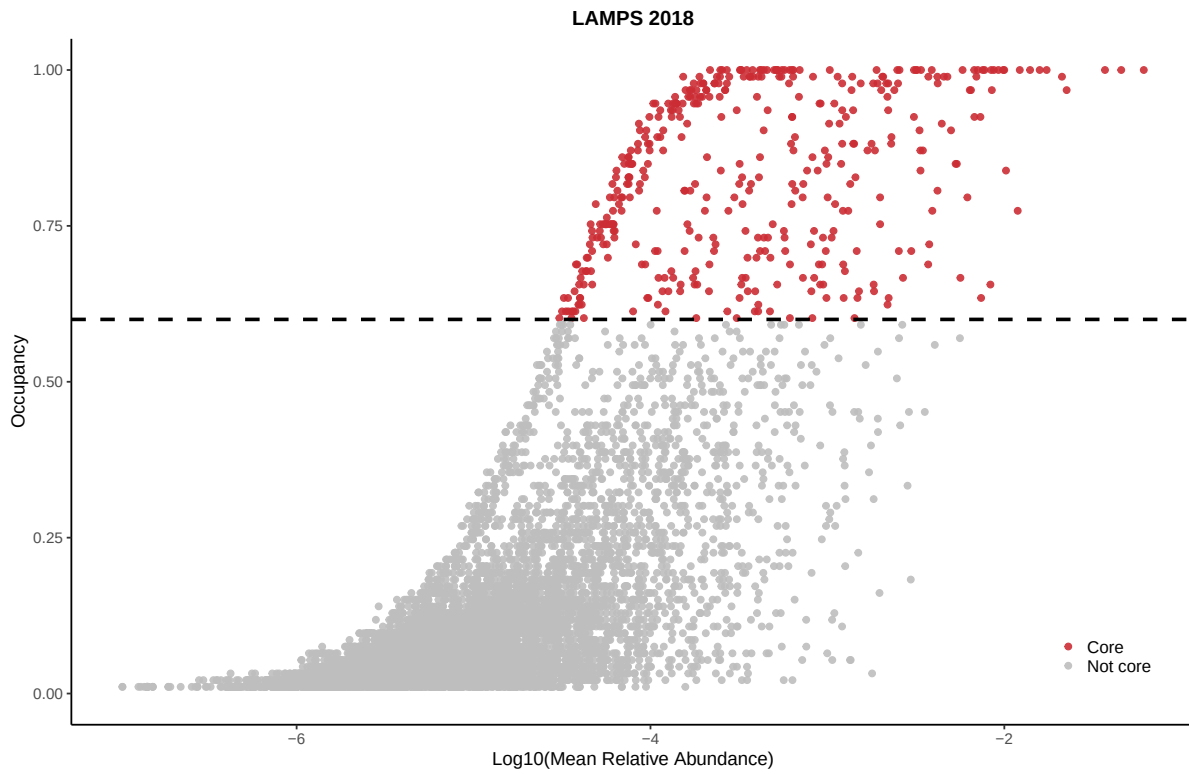
Occupancy threshold: 0.6



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Abundance–Occupancy Curves: Miscanthus ITS– 1 project(s)

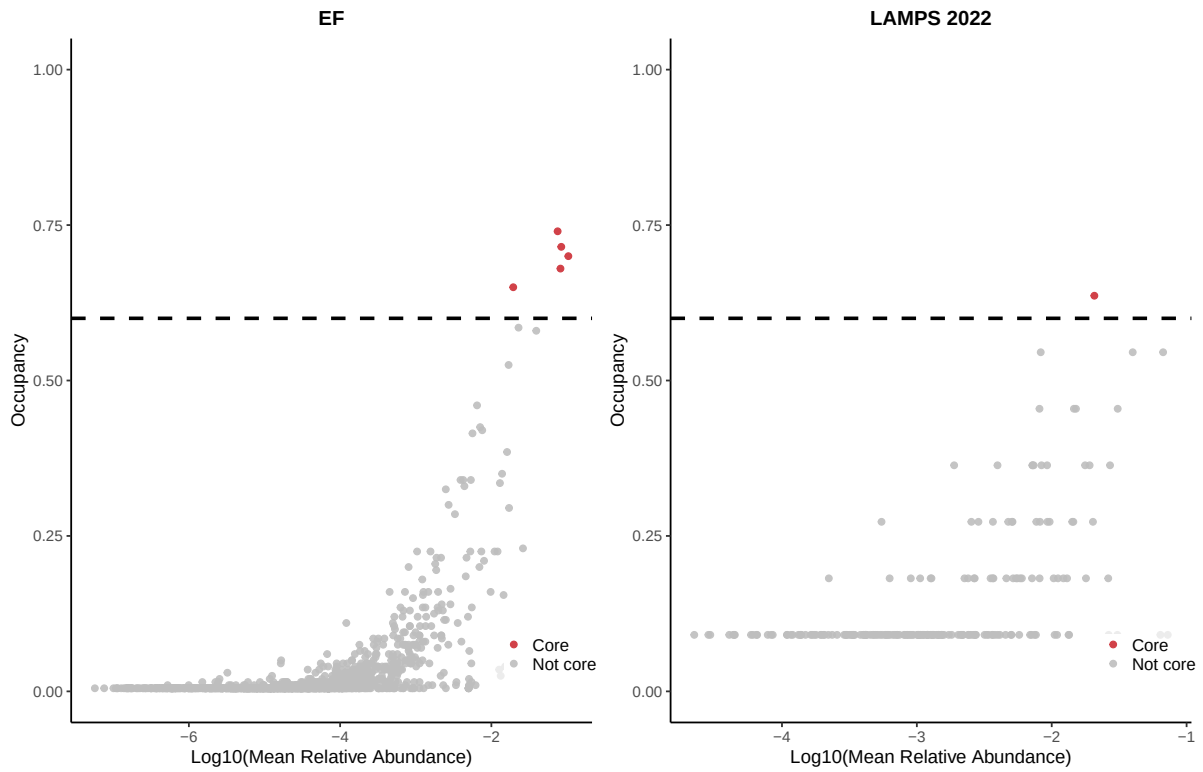
Occupancy threshold: 0.6



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Abundance–Occupancy Curves: Miscanthus AMF– 2 project(s)

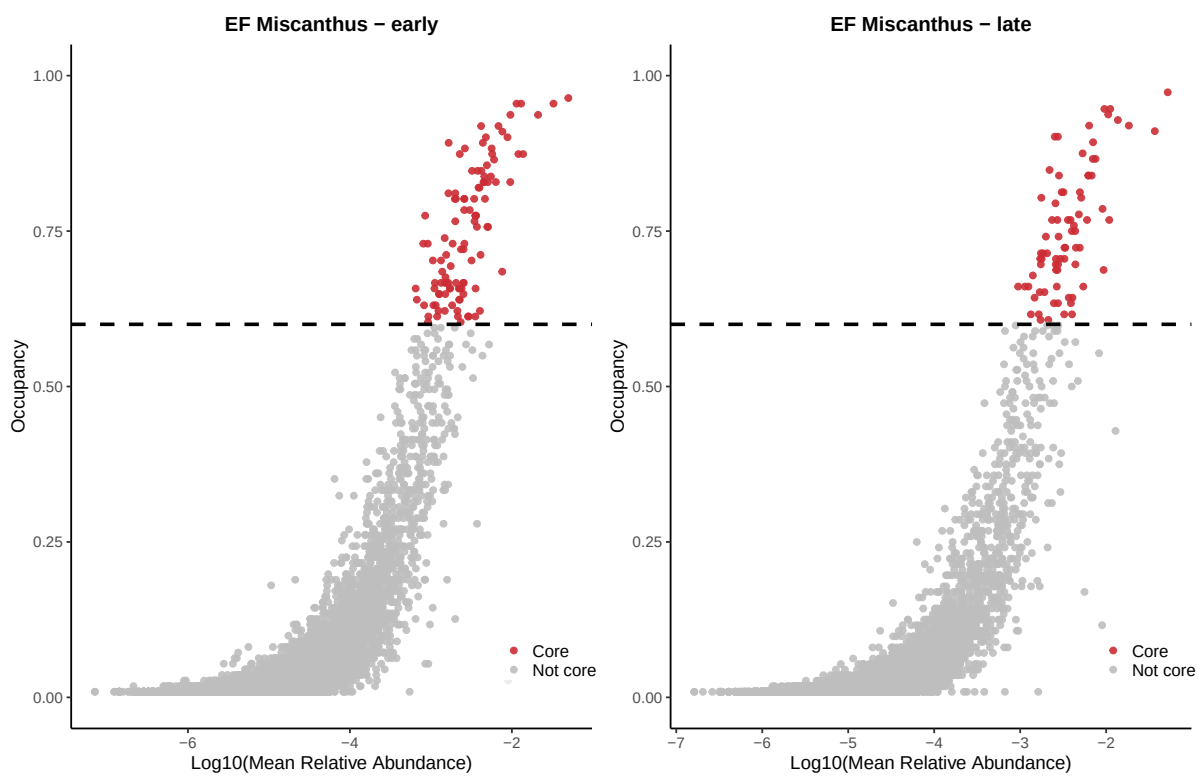
Occupancy threshold: 0.6



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Abundance–Occupancy Curves: Miscanthus 16S – EF by Timing

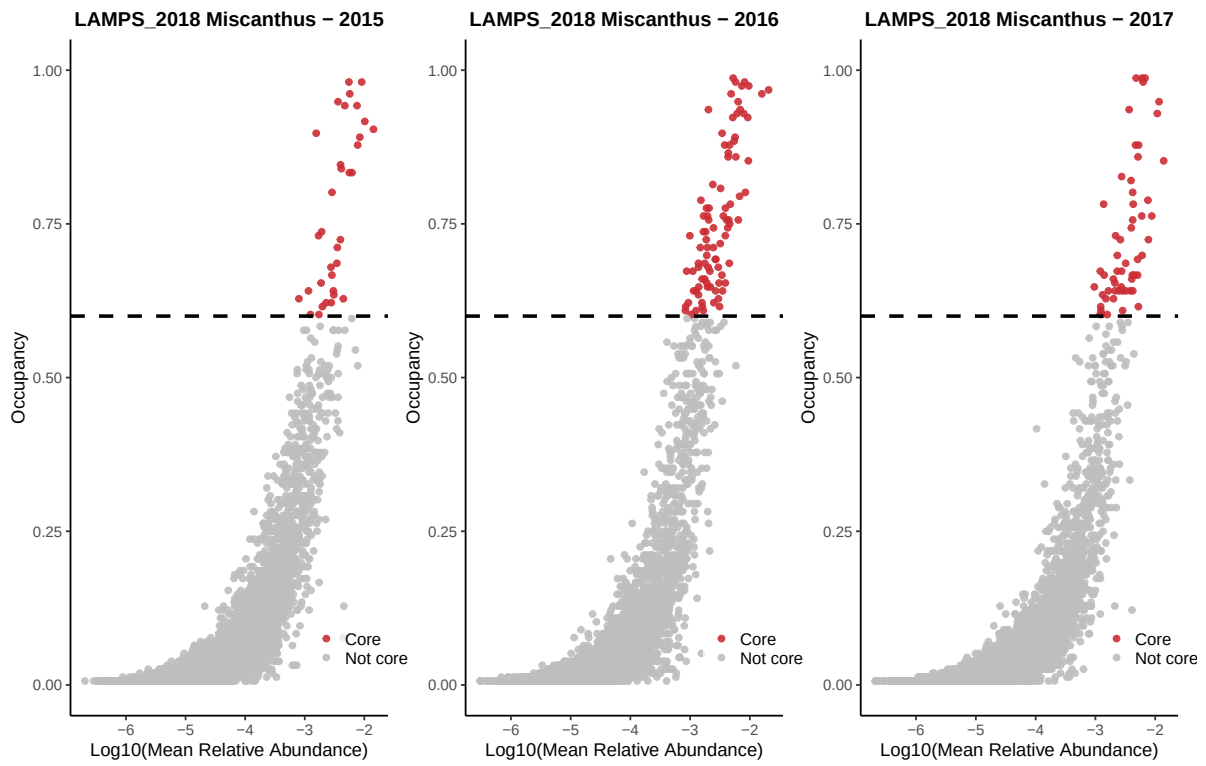
Occupancy threshold: 0.6



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Abundance–Occupancy Curves: Miscanthus 16S – LAMPS 2018 by Year

Occupancy threshold: 0.6

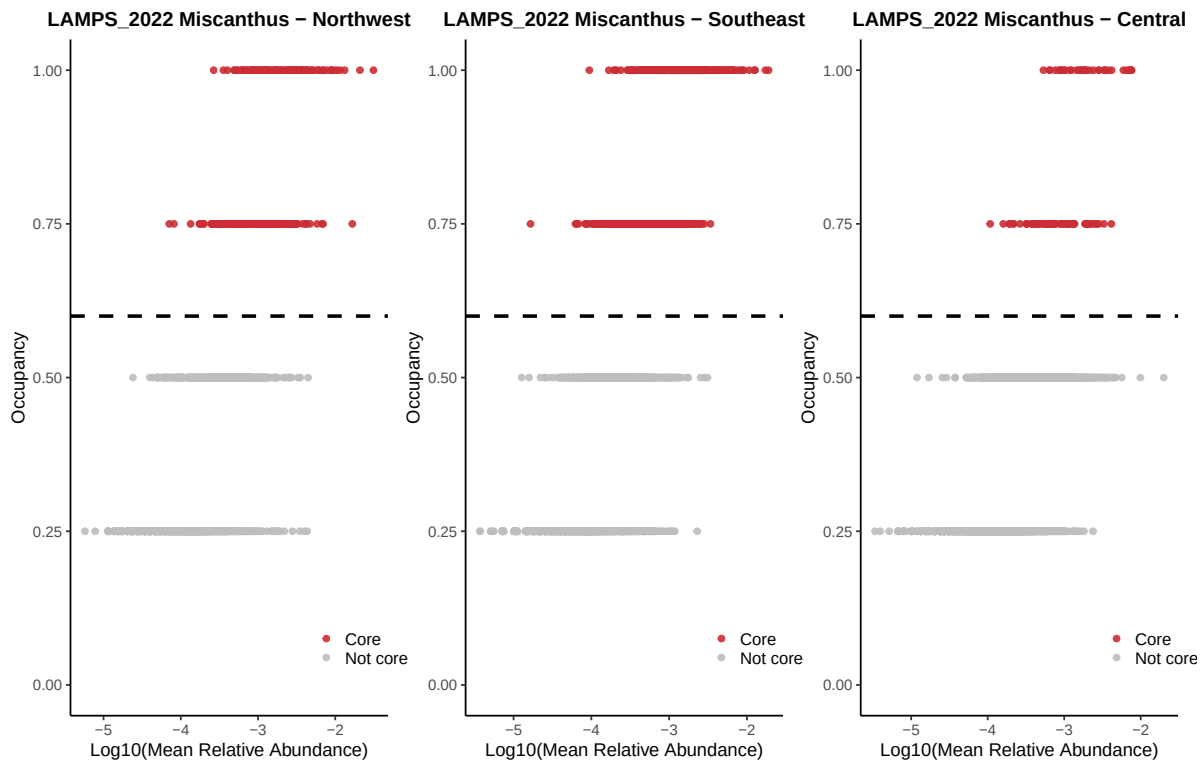


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Abundance–Occupancy Curves: *Miscanthus* 16S – LAMPS 2022 by Site

Occupancy threshold: 0.6

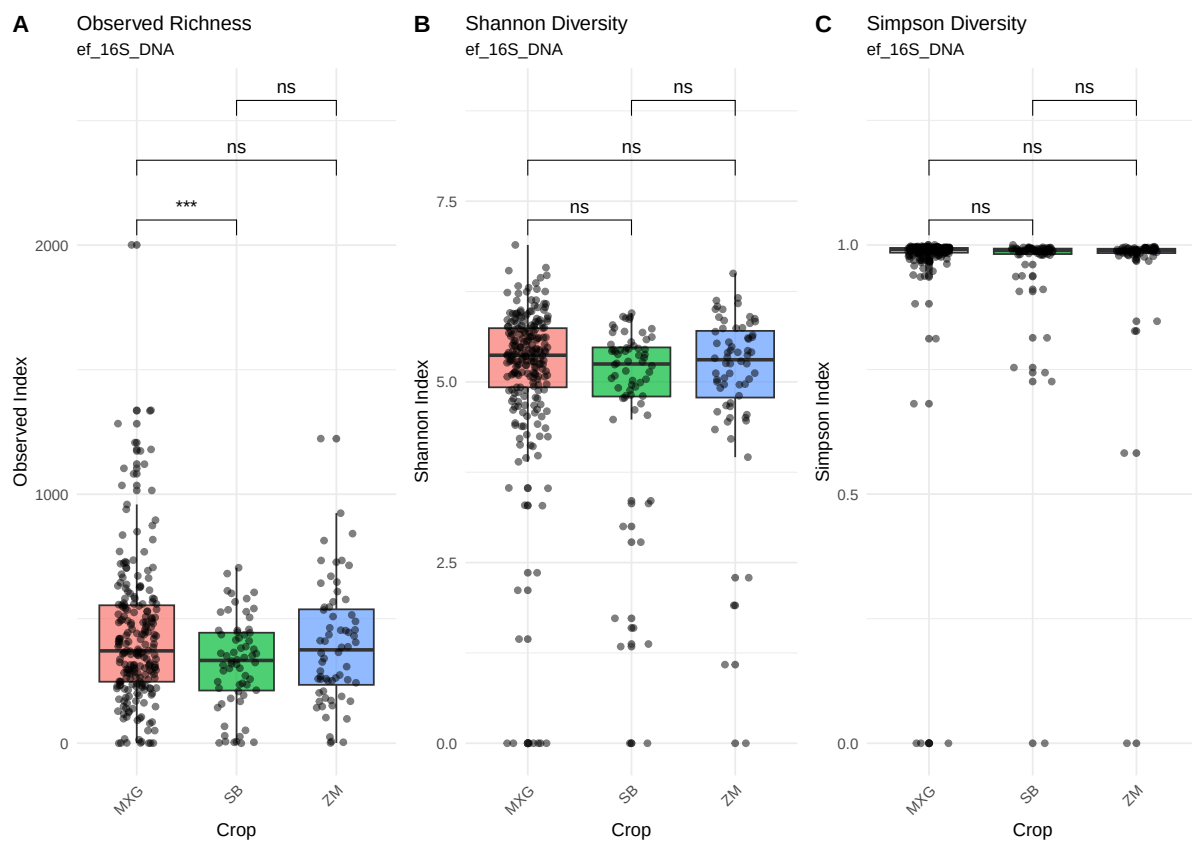


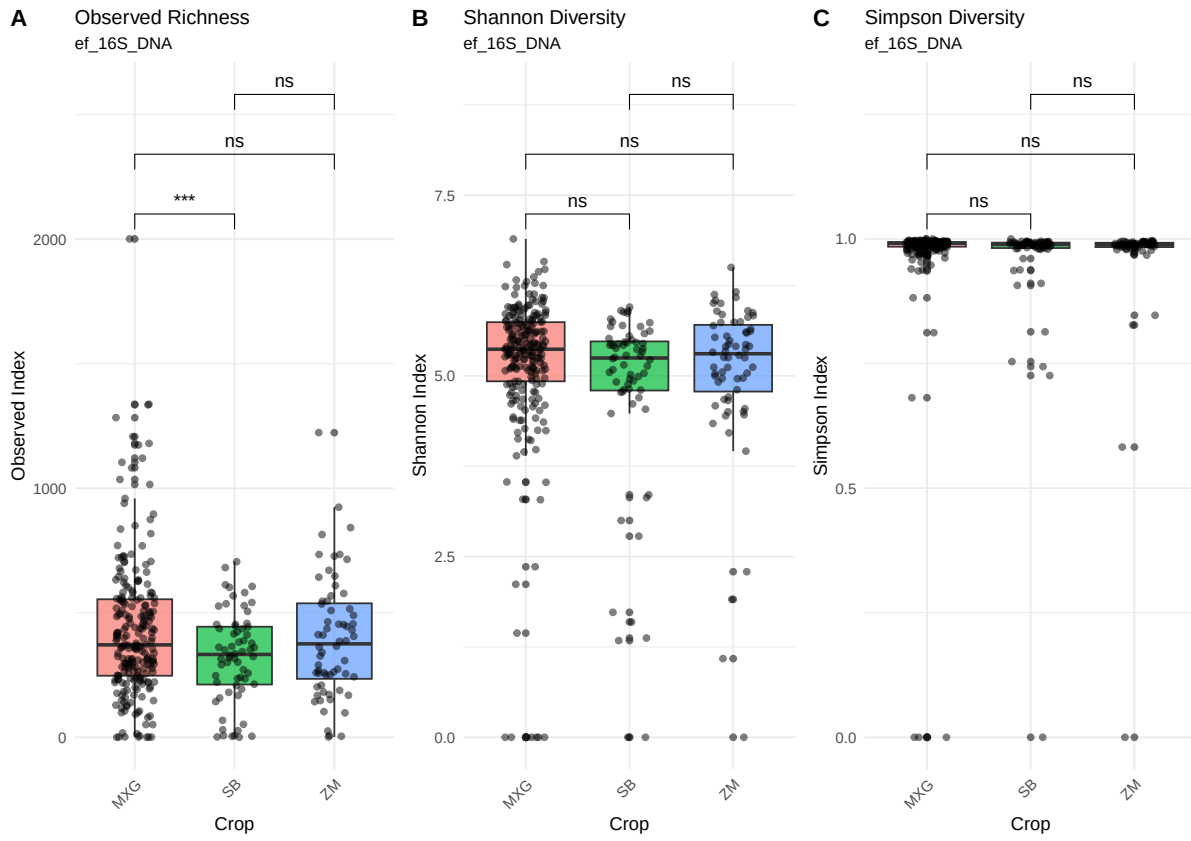
Section 2 — Alpha Diversity Analyses

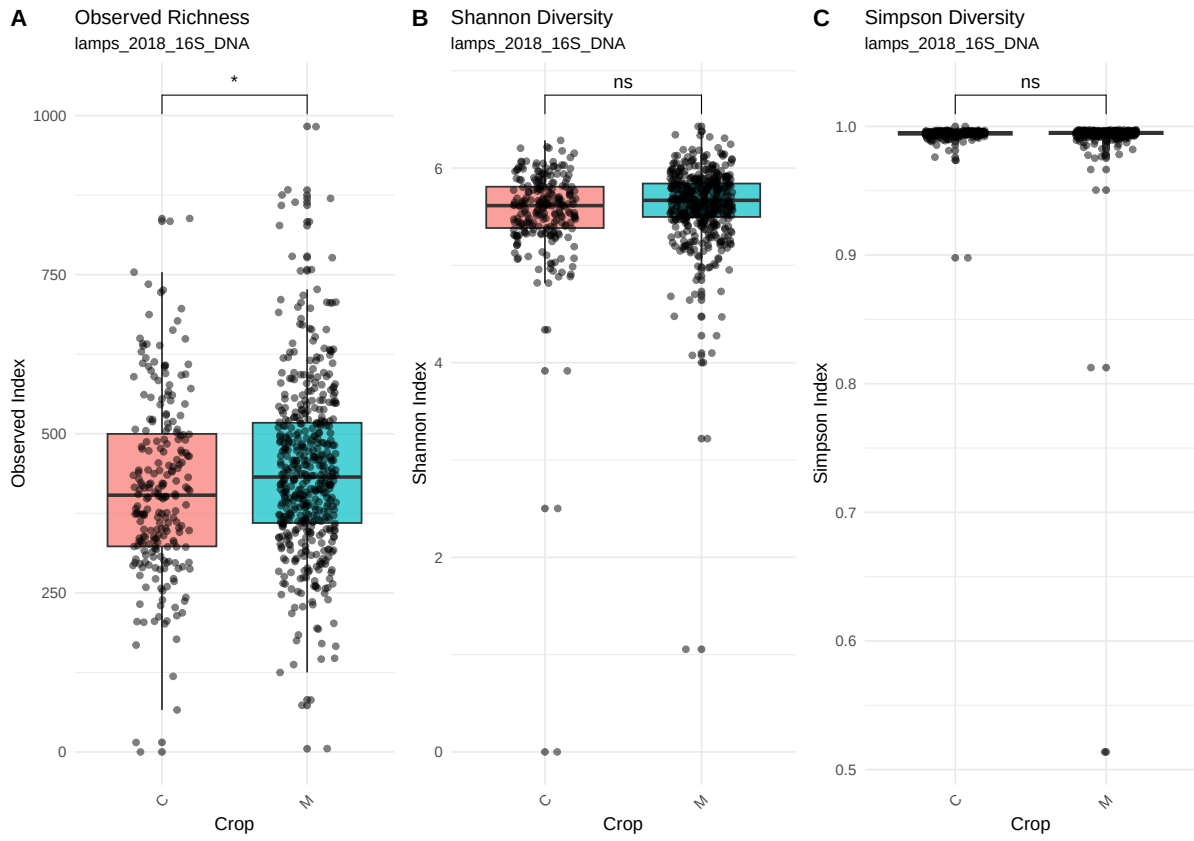
This section runs `06_alpha_beta_div.R`, including:

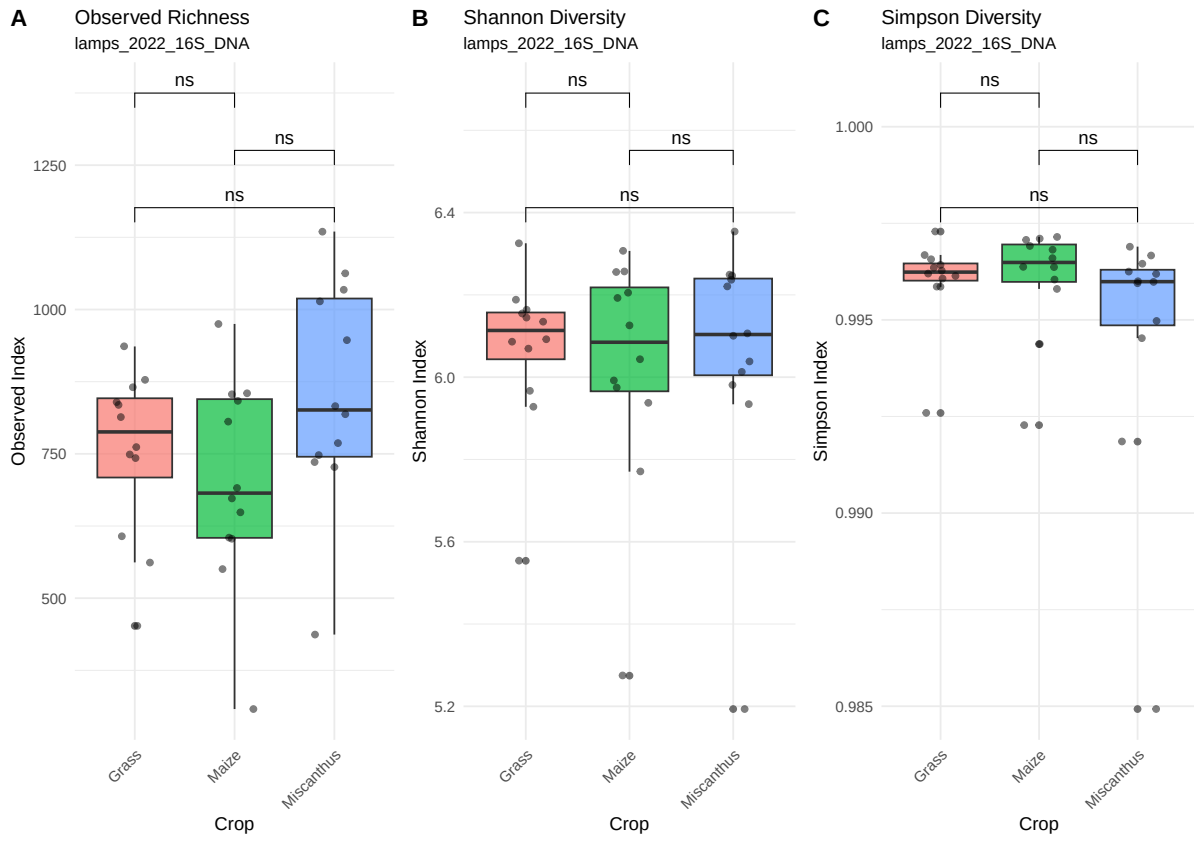
- Richness and alpha diversity summaries
- Beta diversity ordinations

```
source("R/06_alpha_beta_div.R", local = knitr::knit_global())
```

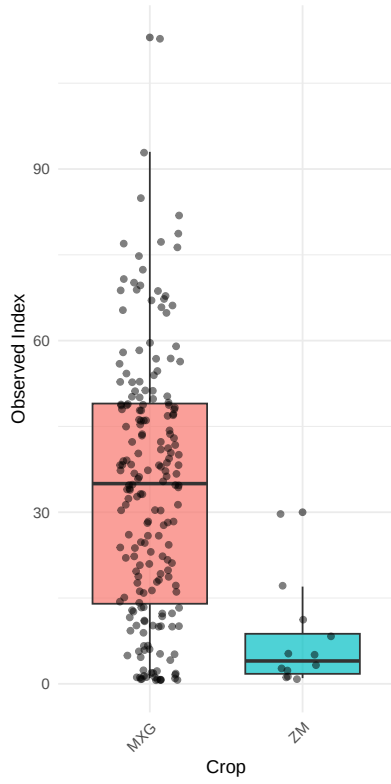





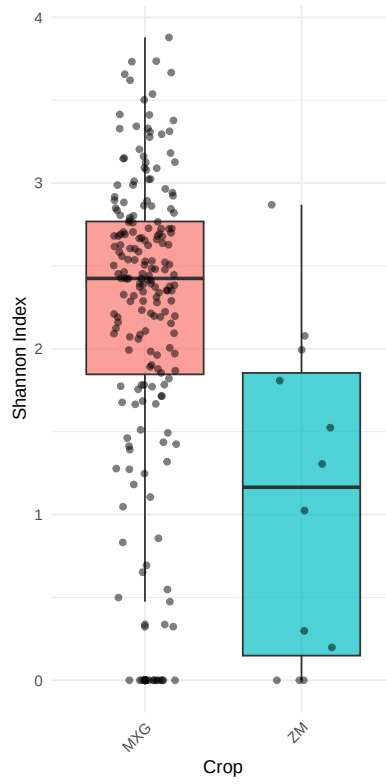




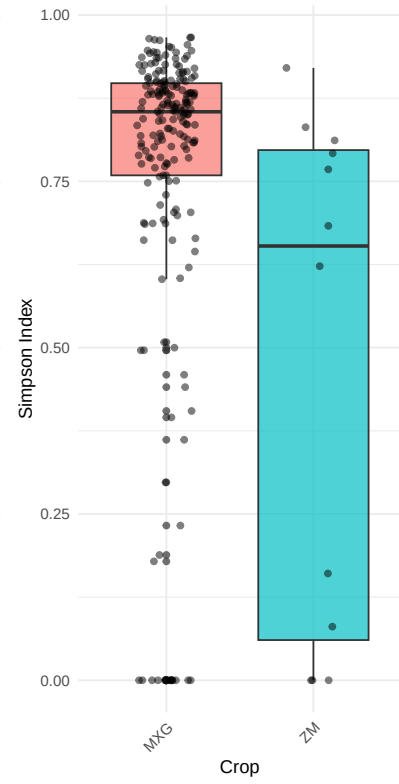
A Observed Richness
ef_AMF_physeq

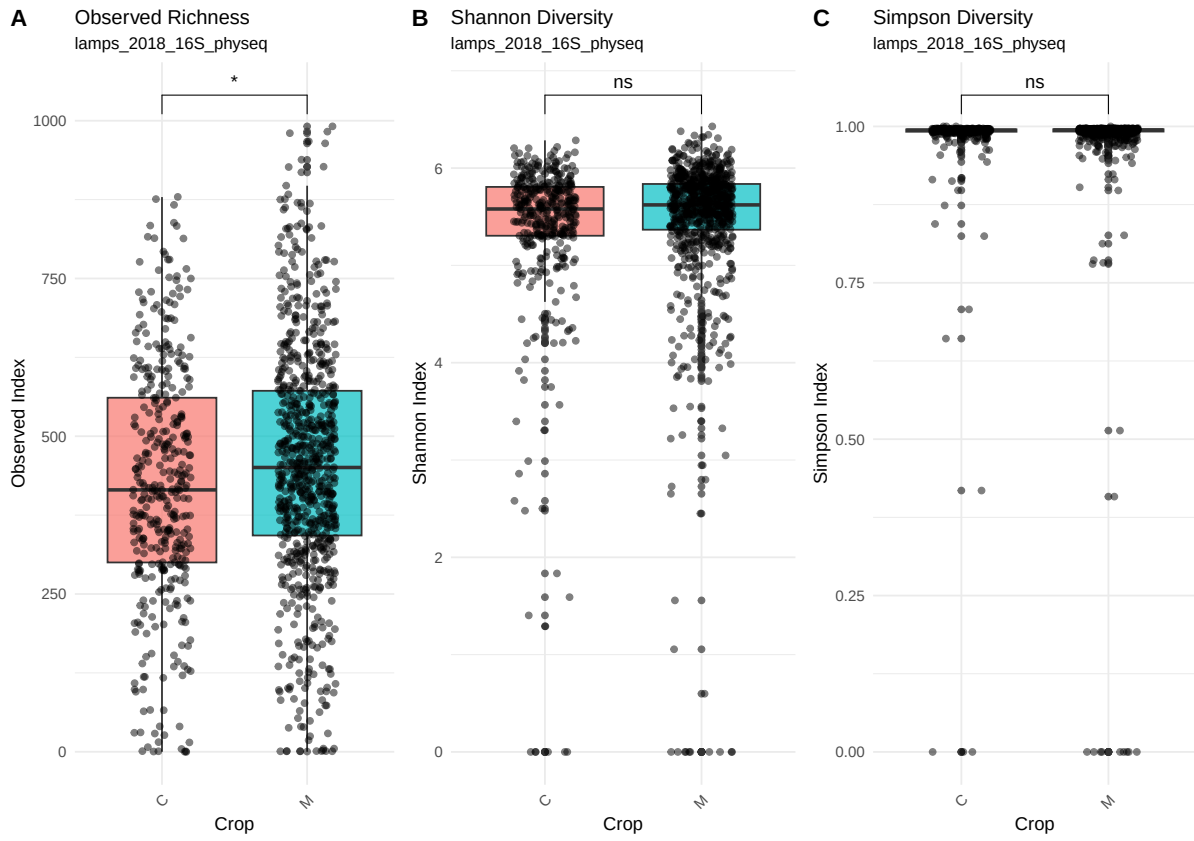


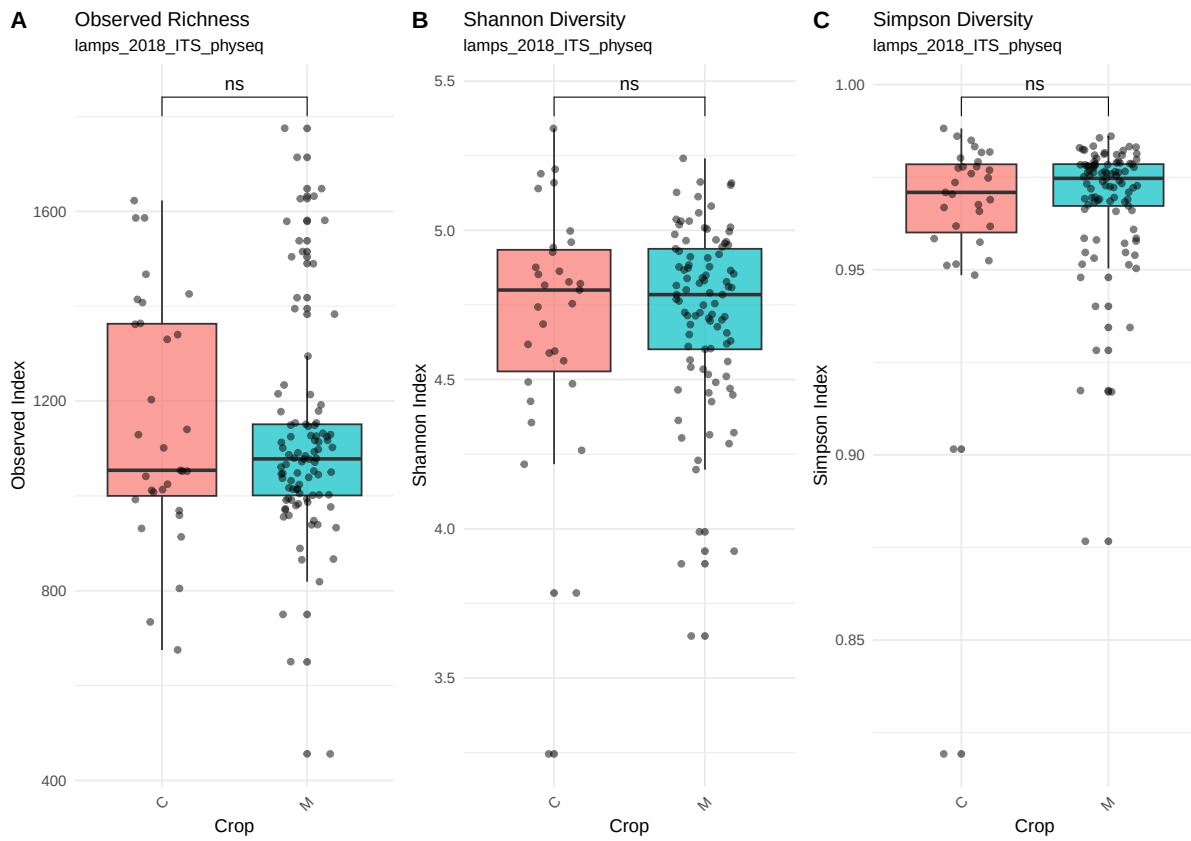
B Shannon Diversity
ef_AMF_physeq

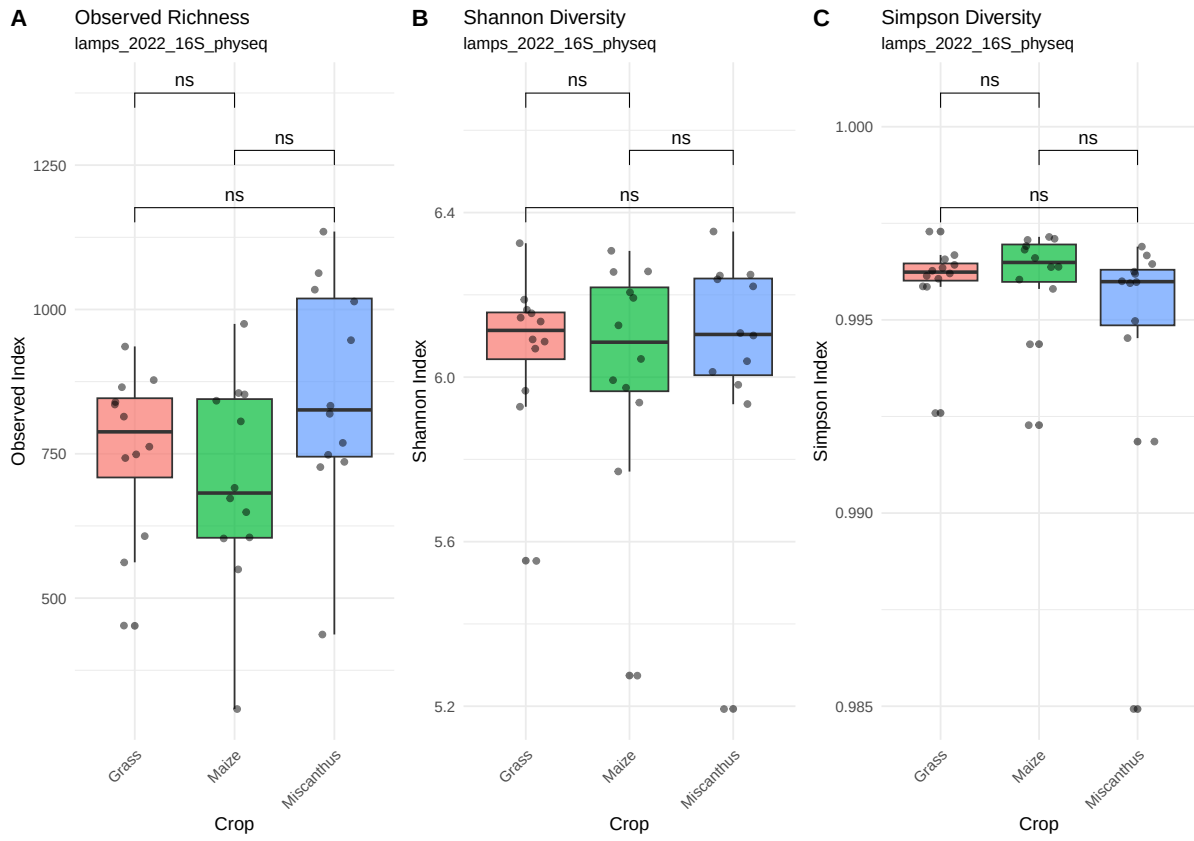


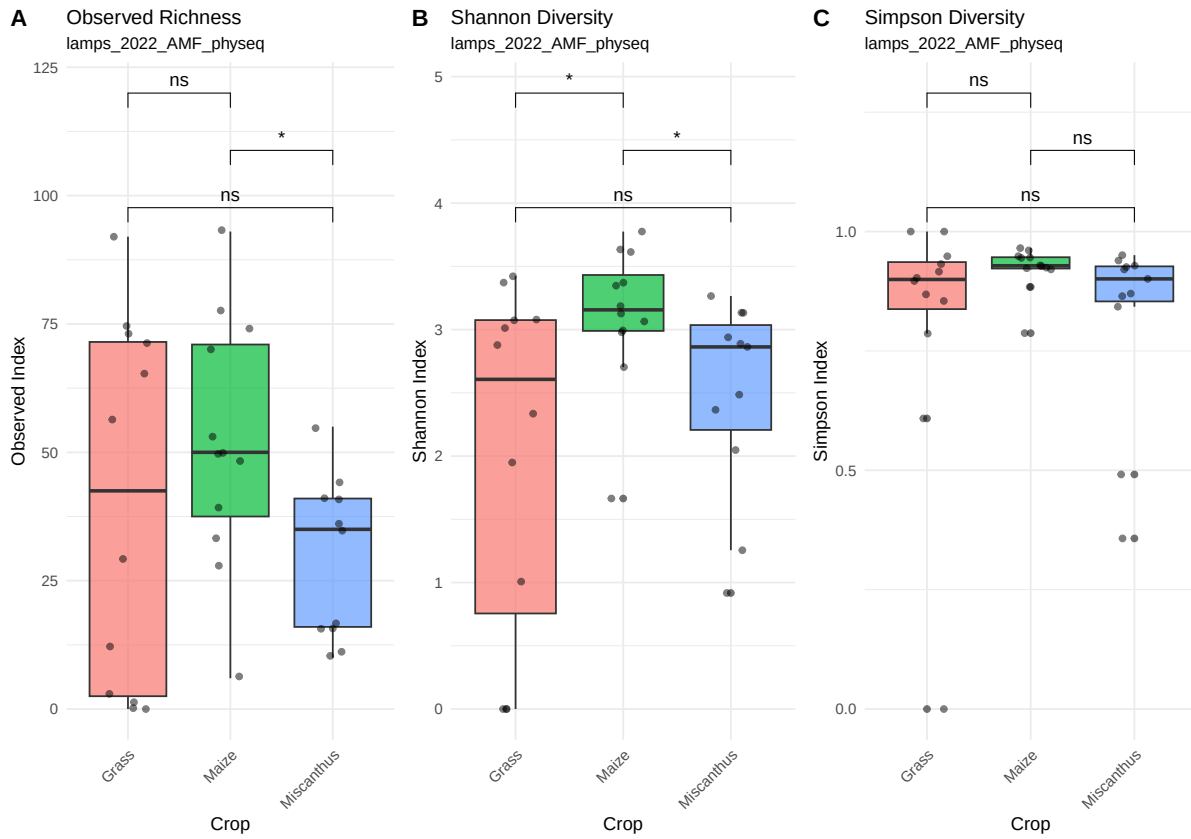
C Simpson Diversity
ef_AMF_physeq

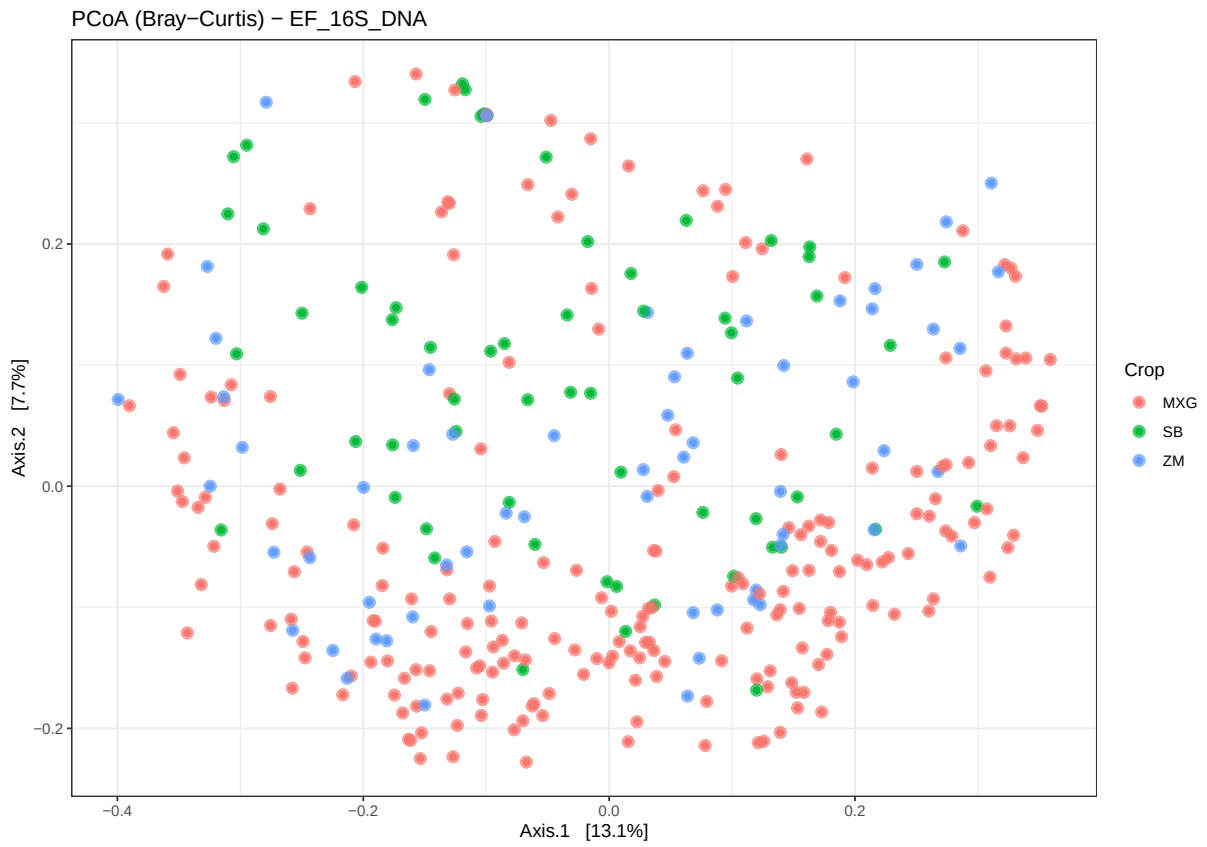


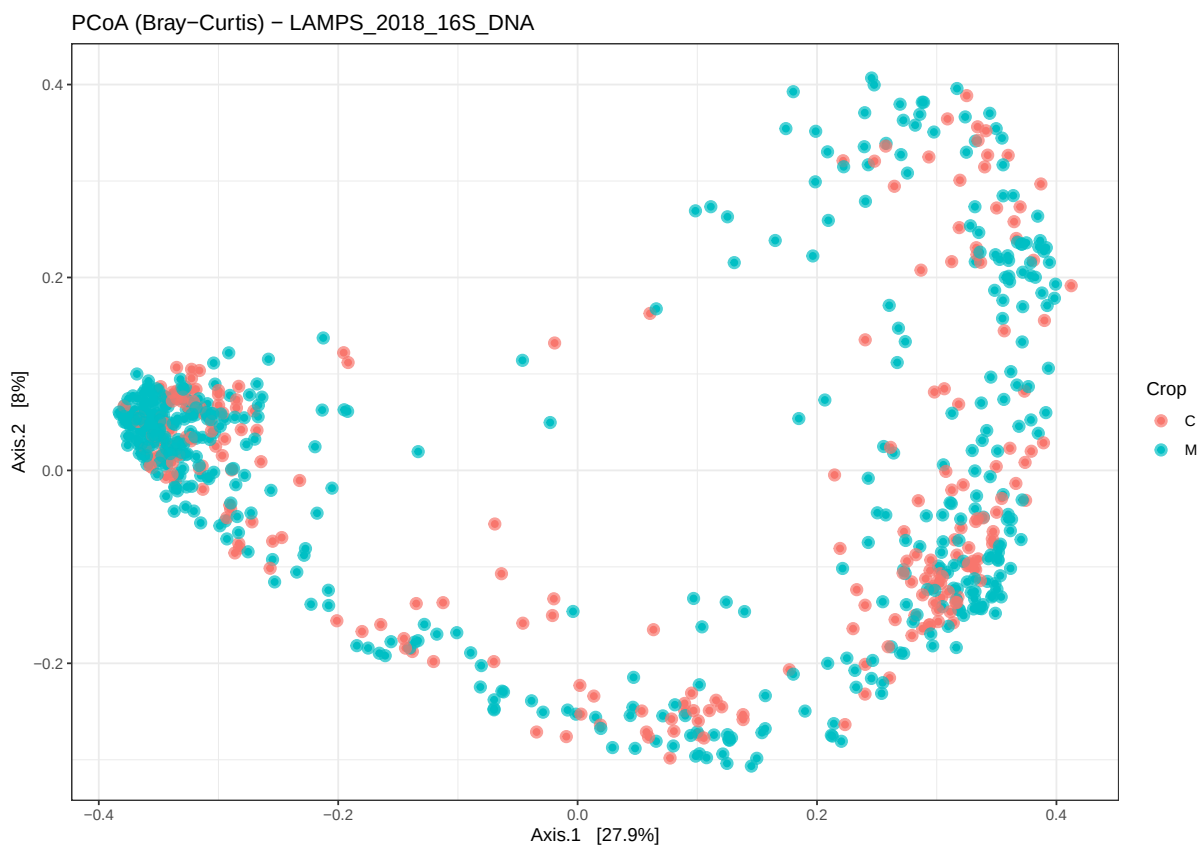


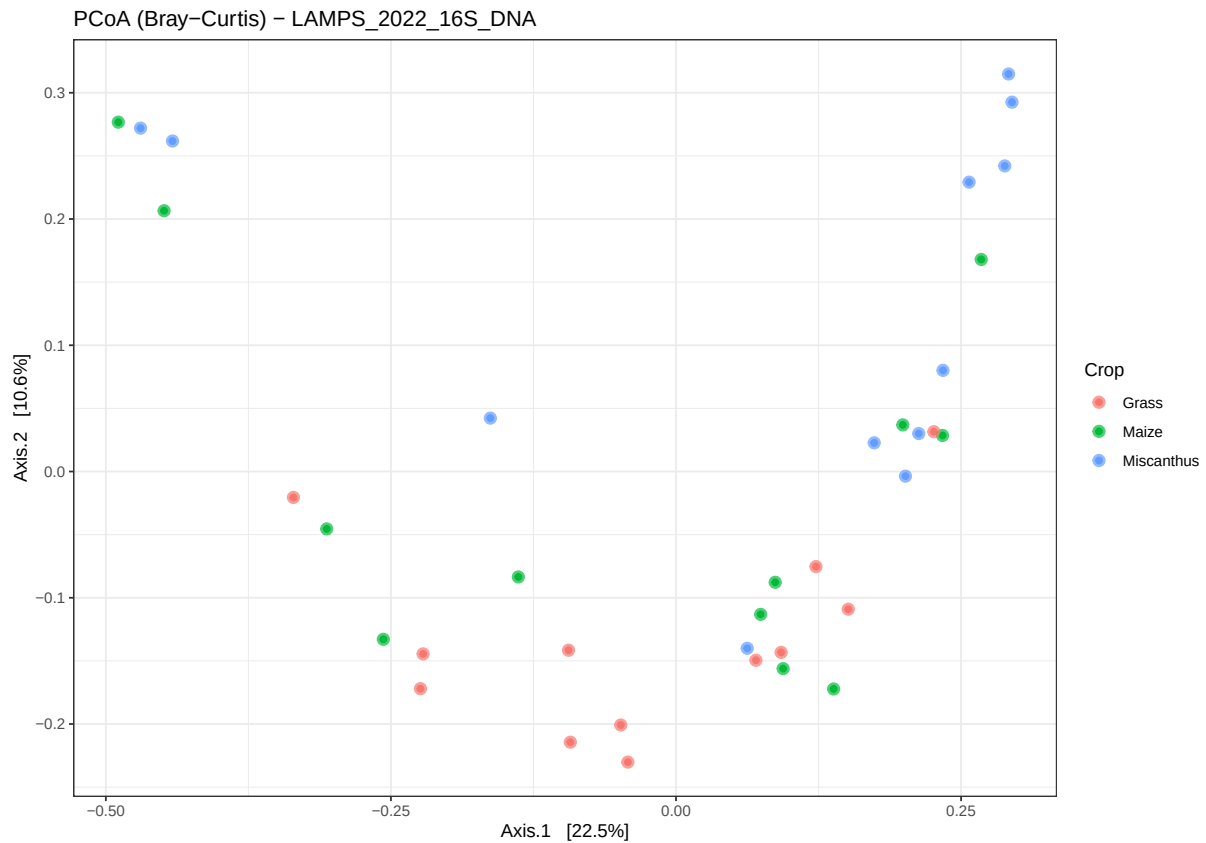












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Alpha Diversity by Project and Crop

=====

\$ef_16S_DNA

\$ef_16S_DNA\$all_div_plots

\$lamps_2018_16S_DNA

\$lamps_2018_16S_DNA\$all_div_plots

\$lamps_2022_16S_DNA

\$lamps_2022_16S_DNA\$all_div_plots

\$ef_physeq_list

\$ef_physeq_list\$ef_16S_physeq

\$ef_physeq_list\$ef_16S_physeq\$all_div_plots

```
$ef_physeq_list$ef_AMF_physeq
$ef_physeq_list$ef_AMF_physeq$all_div_plots
```

```
$lamps_2018_physeq_list
$lamps_2018_physeq_list$lamps_2018_16S_physeq
$lamps_2018_physeq_list$lamps_2018_16S_physeq$all_div_plots
```

```
$lamps_2018_physeq_list$lamps_2018_ITS_physeq
$lamps_2018_physeq_list$lamps_2018_ITS_physeq$all_div_plots
```

```
$lamps_2022_physeq_list
$lamps_2022_physeq_list$lamps_2022_16S_physeq
$lamps_2022_physeq_list$lamps_2022_16S_physeq$all_div_plots
```

```
$lamps_2022_physeq_list$lamps_2022_AMF_physeq
$lamps_2022_physeq_list$lamps_2022_AMF_physeq$all_div_plots
```

```

= = = = =
Alpha Diversity Summary by Project and Crop
= = = = =
# A tibble: 8 x 9
  project crop      n mean_observed sd_observed mean_shannon sd_shannon
  <chr>   <chr> <int>      <dbl>      <dbl>      <dbl>      <dbl>
1 ef     MXG     223      429.       284.       5.13      1.17
2 ef     SB      64      323.       183.       4.75      1.39
3 ef     ZM      64      394.       242.       5.05      1.13
4 lamps  C      228      415.       138.       5.54      0.538
5 lamps  Grass    12      754.       144.       6.07      0.191
6 lamps  M      468      442.       134.       5.62      0.404
7 lamps  Maize    12      701.       179.       6.03      0.286
8 lamps  Miscanthus 12      855.       194.       6.06      0.301
# i 2 more variables: mean_simpson <dbl>, sd_simpson <dbl>

```

```

= = = = =
Beta Diversity Project and Crop
= = = = =
$ef_16S_DNA

$lamps_2018_16S_DNA

$lamps_2022_16S_DNA

```

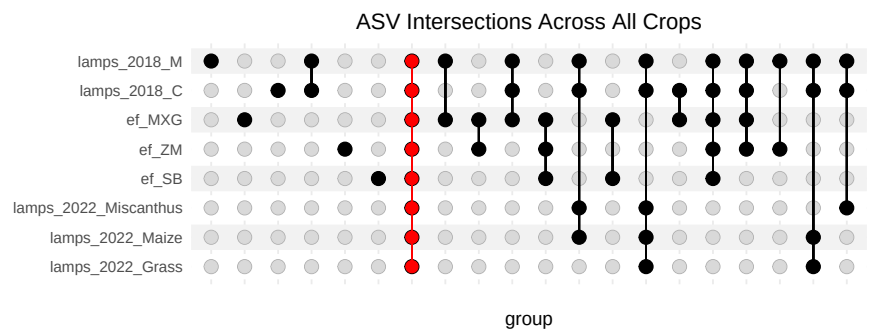
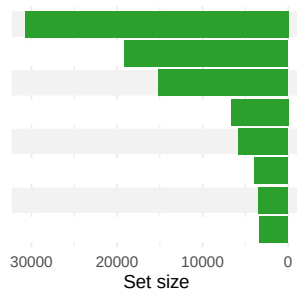
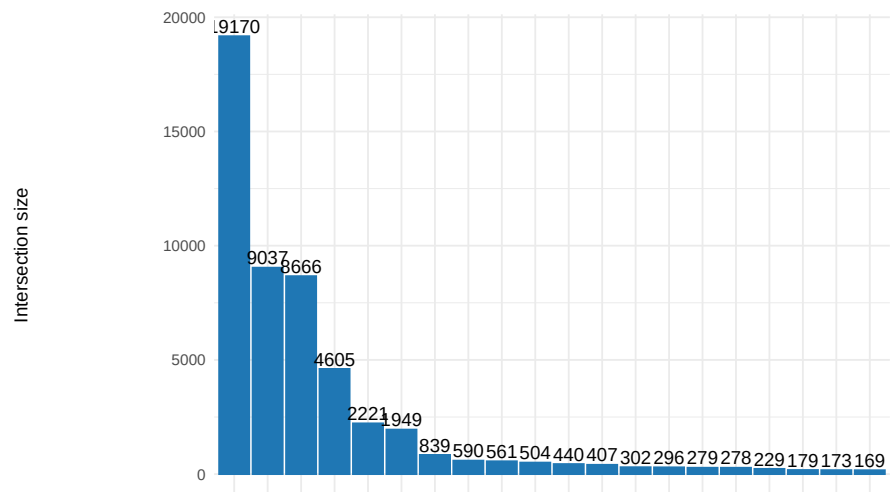
Section 3 — Shared ASVs Analyses

This section runs [08_shared_asvs.R](#), including: - Upset plots of shared ASVs across projects, regions, and sample types - Summary tables of shared ASVs at different prevalence thresholds

```

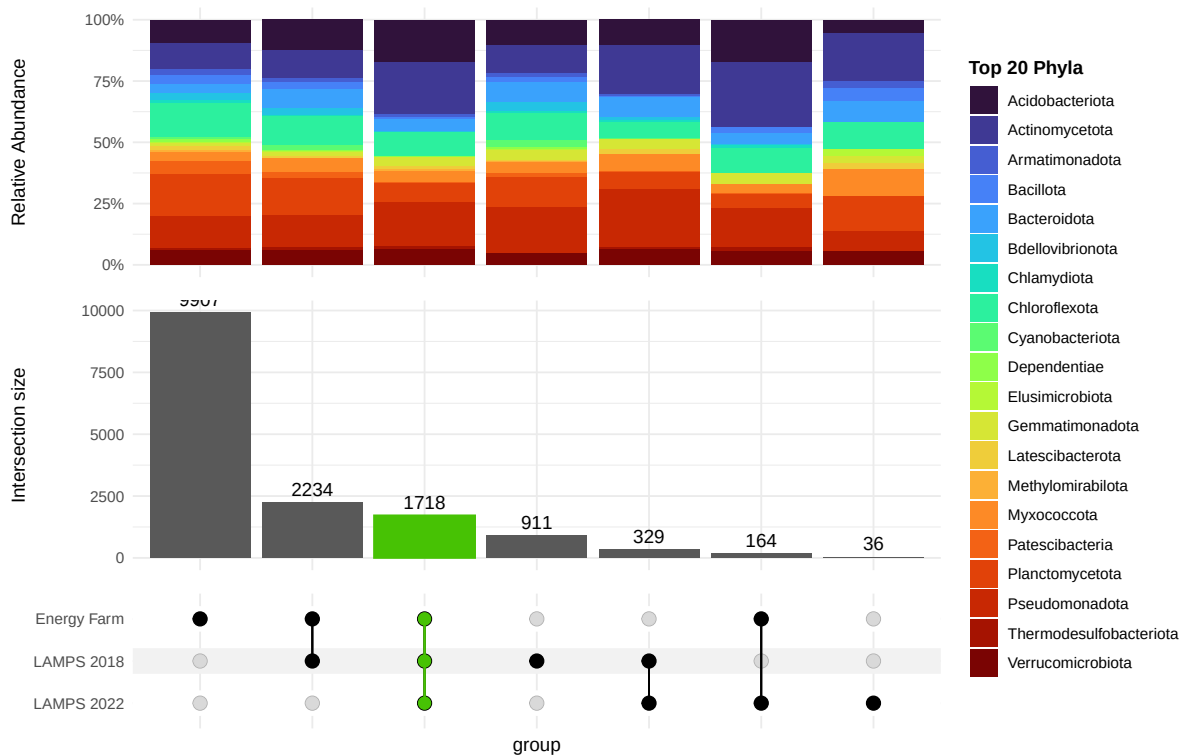
source(
  "R/08_shared_asvs_upset_plots.R",
  local = knitr::knit_global(),
  echo = FALSE,
  print.eval = TRUE
)

```

ASVs in Top 20 Phyla in Miscanthus

16S data only



Taxonomic Richness by Crop-Project

A tibble: 8 x 4

crop_project	n_asvs	n_phyla	n_classes
<chr>	<int>	<int>	<int>
1 lamps_2018_M	30769	54	117
2 lamps_2018_C	19166	54	112
3 ef_MXG	15182	51	114
4 ef_ZM	6678	42	98
5 ef_SB	5809	41	95
6 lamps_2022_Miscanthus	3939	37	87
7 lamps_2022_Maize	3503	36	86
8 lamps_2022_Grass	3336	37	85

ASV Intersections Across All Crops

```

=====
ASV Intersections - Miscanthus Only
=====

=====
ASV Sharing Summary
=====
# A tibble: 3 x 3
  n_sets count category
  <dbl> <int> <chr>
1     1  10854 Unique to one crop
2     2   2727 Shared by 2 crops
3     3   1718 Shared by 3 crops

=====
Shared ASVs in Miscanthus
=====
# A tibble: 20 x 3
  phylum          n    pct
  <chr>         <int> <dbl>
1 Actinomycetota     364 21.2
2 Pseudomonadota     310 18.0
3 Acidobacteriota    299 17.4
4 Chloroflexota      168  9.78
5 Planctomycetota    130  7.57
6 Verrucomicrobiota  112  6.52
7 Bacteroidota       84  4.89
8 Myxococcota        77  4.48
9 Gemmatimonadota    61  3.55
10 Latescibacterota   23  1.34
11 Armatimonadota     22  1.28
12 Thermodesulfobacteriota 17  0.990
13 Bacillota          14  0.815
14 Methyloirabilota   13  0.757
15 Patescibacteria     9  0.524
16 Cyanobacteriota     4  0.233
17 Elusimicrobiota     4  0.233
18 Bdellovibrionota    3  0.175
19 Chlamydiota         2  0.116
20 Dependitiae         2  0.116

=====

```

Comparison of Shared vs Unique ASVs by Phylum:

A tibble: 20 x 4

phylum	`Core (all 3)`	`Shared (2)`	`Unique (1)`
<chr>	<int>	<int>	<int>
1 Acidobacteriota	299	337	1063
2 Actinomycetota	364	367	1141
3 Armatimonadota	22	42	275
4 Bacillota	14	70	347
5 Bacteroidota	84	202	471
6 Bdellovibrionota	3	70	309
7 Chlamydiota	2	17	139
8 Chloroflexota	168	304	1451
9 Cyanobacteriota	4	48	137
10 Dependitiae	2	4	113
11 Elusimicrobiota	4	9	61
12 Gemmatimonadota	61	56	166
13 Latescibacterota	23	23	136
14 Methyloirabilota	13	8	82
15 Myxococcota	77	152	434
16 Patescibacteria	9	58	510
17 Planctomycetota	130	368	1850
18 Pseudomonadota	310	395	1450
19 Thermodesulfobacteriota	17	33	85
20 Verrucomicrobiota	112	164	634

= = = = =

Details of Shared ASVs:

A tibble: 1,718 x 6

asv	n_sets	sets_present	phylum	class	genus
<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>
1 TACGTAGGGCGCAAGCGTTATCCGGAATTATTGGGCG~	3	ef_MXG, lam~	Actin~	Acti~	Arth~
2 TACAGAGGTCTCAAGCGTTGTTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Verru~	Verr~	Cand~
3 TACAGAGGTCTCAAGCGTTGTTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Verru~	Verr~	Cand~
4 TACAGAGGTCTCAAGCGTTGTTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Verru~	Verr~	Cand~
5 TACGTAGGGGGCAAGCGTTGTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Actin~	Ther~	<NA>
6 TACAGAGGTCTCAAGCGTTGTTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Verru~	Verr~	Cand~
7 TACAGAGGTCCCAAGCGTTGTTCCGATTTCATTGGGCG~	3	ef_MXG, lam~	Verru~	Verr~	Cand~
8 TACAGAGGGGGCAAGCGTTGTTCCGGAATTACTGGGCG~	3	ef_MXG, lam~	Acido~	Vici~	<NA>
9 GACAGAGGGGGCAAGCGTTGTTCCGGAATTACTGGGCG~	3	ef_MXG, lam~	Methy~	Meth~	<NA>
10 TACGGGGGGGGCAAGCGTTGTTCCGGAATTACTGGGCG~	3	ef_MXG, lam~	Acido~	Vici~	<NA>

i 1,708 more rows

= = = = =

Shared ASVs Between Projects - PROJECT LEVEL Summary

= = = = =

Threshold: threshold_0

ASVs above threshold shared across ALL projects: 1794

ASVs above threshold shared by 2+ projects: 5871

ASVs above threshold by project:

ef_16S : 15182 ASVs

lamps_2018_16S : 30769 ASVs

lamps_2022_16S : 3939 ASVs

Threshold: threshold_0.6

ASVs above threshold shared across ALL projects: 9

ASVs above threshold shared by 2+ projects: 58

ASVs above threshold by project:

ef_16S : 88 ASVs

lamps_2018_16S : 32 ASVs

lamps_2022_16S : 211 ASVs

Threshold: threshold_0.7

ASVs above threshold shared across ALL projects: 6

ASVs above threshold shared by 2+ projects: 39

ASVs above threshold by project:

ef_16S : 56 ASVs

lamps_2018_16S : 20 ASVs

lamps_2022_16S : 133 ASVs

Threshold: threshold_0.8

ASVs above threshold shared across ALL projects: 4

ASVs above threshold shared by 2+ projects: 15

ASVs above threshold by project:

ef_16S : 28 ASVs

lamps_2018_16S : 14 ASVs

lamps_2022_16S : 73 ASVs

Threshold: threshold_0.9

ASVs above threshold shared across ALL projects: 1

ASVs above threshold shared by 2+ projects: 7

ASVs above threshold by project:

ef_16S : 8 ASVs

lamps_2018_16S : 6 ASVs

lamps_2022_16S : 35 ASVs

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Shared ASVs Between Crops - CROP LEVEL Summary

=====

Threshold: threshold_0

ASVs above threshold shared across ALL crops: 839

ASVs above threshold shared by 2+ crops: 14281

ASVs above threshold by crop (showing first 10):

ef_16S_MXG : 15182 ASVs
 ef_16S_SB : 5809 ASVs
 ef_16S_ZM : 6678 ASVs
 lamps_2018_16S_C : 19166 ASVs
 lamps_2018_16S_M : 30769 ASVs
 lamps_2022_16S_Maize : 3503 ASVs
 lamps_2022_16S_Miscanthus : 3939 ASVs
 lamps_2022_16S_Grass : 3336 ASVs

Threshold: threshold_0.6

ASVs above threshold shared across ALL crops: 9

ASVs above threshold shared by 2+ crops: 184

ASVs above threshold by crop (showing first 10):

ef_16S_MXG : 88 ASVs
 ef_16S_SB : 69 ASVs
 ef_16S_ZM : 85 ASVs
 lamps_2018_16S_C : 34 ASVs
 lamps_2018_16S_M : 32 ASVs
 lamps_2022_16S_Maize : 133 ASVs
 lamps_2022_16S_Miscanthus : 211 ASVs
 lamps_2022_16S_Grass : 209 ASVs

Threshold: threshold_0.7

ASVs above threshold shared across ALL crops: 6

ASVs above threshold shared by 2+ crops: 117

ASVs above threshold by crop (showing first 10):

ef_16S_MXG : 56 ASVs
 ef_16S_SB : 48 ASVs
 ef_16S_ZM : 59 ASVs
 lamps_2018_16S_C : 23 ASVs
 lamps_2018_16S_M : 20 ASVs
 lamps_2022_16S_Maize : 83 ASVs
 lamps_2022_16S_Miscanthus : 133 ASVs
 lamps_2022_16S_Grass : 143 ASVs

Threshold: threshold_0.8

ASVs above threshold shared across ALL crops: 3

ASVs above threshold shared by 2+ crops: 72

ASVs above threshold by crop (showing first 10):

ef_16S_MXG : 28 ASVs

ef_16S_SB : 21 ASVs

ef_16S_ZM : 28 ASVs

lamps_2018_16S_C : 15 ASVs

lamps_2018_16S_M : 14 ASVs

lamps_2022_16S_Maize : 48 ASVs

lamps_2022_16S_Miscanthus : 73 ASVs

lamps_2022_16S_Grass : 106 ASVs

Threshold: threshold_0.9

ASVs above threshold shared across ALL crops: 1

ASVs above threshold shared by 2+ crops: 39

ASVs above threshold by crop (showing first 10):

ef_16S_MXG : 8 ASVs

ef_16S_SB : 1 ASVs

ef_16S_ZM : 10 ASVs

lamps_2018_16S_C : 6 ASVs

lamps_2018_16S_M : 6 ASVs

lamps_2022_16S_Maize : 29 ASVs

lamps_2022_16S_Miscanthus : 35 ASVs

lamps_2022_16S_Grass : 67 ASVs

Session information

```
sessionInfo()
```

R version 4.4.2 (2024-10-31)

Platform: x86_64-pc-linux-gnu

Running under: Arch Linux

Matrix products: default

BLAS/LAPACK: /usr/lib/libopenblas.so.0.3; LAPACK version 3.12.0

locale:

[1] LC_CTYPE=C.UTF-8

LC_NUMERIC=C

LC_TIME=C.UTF-8

```

[4] LC_COLLATE=C.UTF-8      LC_MONETARY=C.UTF-8      LC_MESSAGES=C.UTF-8
[7] LC_PAPER=C.UTF-8        LC_NAME=C                 LC_ADDRESS=C
[10] LC_TELEPHONE=C          LC_MEASUREMENT=C.UTF-8   LC_IDENTIFICATION=C

```

```

time zone: America/Chicago
tzcode source: system (glibc)

```

attached base packages:

```

[1] stats      graphics  grDevices datasets  utils      methods    base

```

other attached packages:

```

[1] ComplexUpset_1.3.3  gtExtras_0.6.2      gt_1.3.0
[4] iNEXT_3.0.2         readxl_1.4.5        ggtext_0.1.2
[7] metagMisc_0.6.0.9000 microbiome_1.28.0    janitor_2.2.1
[10] data.table_1.17.8   lubridate_1.9.4      forcats_1.0.1
[13] stringr_1.6.0       dplyr_1.1.4         purrr_1.2.0
[16] readr_2.1.6         tidyr_1.3.1         tibble_3.3.0
[19] ggplot2_3.5.2       tidyverse_2.0.0     vegan_2.7-1
[22] permute_0.9-8       phyloseq_1.50.0     conflicted_1.2.0

```

loaded via a namespace (and not attached):

```

[1] RColorBrewer_1.1-3  jsonlite_2.0.0      magrittr_2.0.4
[4] farver_2.1.2        rmarkdown_2.29      fs_1.6.6
[7] zlibbioc_1.52.0     ragg_1.5.0          vctrs_0.6.5
[10] multtest_2.62.0     memoise_2.0.1       paletteer_1.7.0
[13] rstatix_0.7.3       tinytex_0.54        htmltools_0.5.8.1
[16] DEoptim_2.2-8       broom_1.0.10        cellranger_1.1.0
[19] Rhdf5lib_1.28.0     Formula_1.2-5       rhdf5_2.50.2
[22] fontawesome_0.5.3   plyr_1.8.9          cachem_1.1.0
[25] igraph_2.1.4        lifecycle_1.0.4     iterators_1.0.14
[28] pkgconfig_2.0.3     Matrix_1.7-2        R6_2.6.1
[31] fastmap_1.2.0       GenomeInfoDbData_1.2.13 snakecase_0.11.1
[34] digest_0.6.39       numDeriv_2016.8-1.1 colorspace_2.1-1
[37] rematch2_2.1.2      patchwork_1.3.2     S4Vectors_0.44.0
[40] rprojroot_2.1.1     textshaping_1.0.3   ggpubr_0.6.2
[43] labeling_0.4.3      clusterGeneration_1.3.8 timechange_0.3.0
[46] abind_1.4-8         httr_1.4.7          mgcv_1.9-1
[49] compiler_4.4.2      here_1.0.2          withr_3.0.2
[52] doParallel_1.0.17   S7_0.2.0            backports_1.5.0
[55] optimParallel_1.0-2 carData_3.0-5        maps_3.4.3
[58] ggsignif_0.6.4      MASS_7.3-64         scatterplot3d_0.3-44
[61] biomformat_1.34.0   tools_4.4.2         ape_5.8-1
[64] phytools_2.5-2      glue_1.8.0          quadprog_1.5-8

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[67]	nlme_3.1-167	rhdf5filters_1.18.1	gridtext_0.1.5
[70]	grid_4.4.2	Rtsne_0.17	cluster_2.1.8
[73]	reshape2_1.4.4	ade4_1.7-23	generics_0.1.4
[76]	gtable_0.3.6	tzdb_0.5.0	hms_1.1.4
[79]	utf8_1.2.6	car_3.1-3	xml2_1.5.1
[82]	XVector_0.46.0	BiocGenerics_0.52.0	foreach_1.5.2
[85]	pillar_1.11.1	splines_4.4.2	lattice_0.22-6
[88]	renv_1.1.5	survival_3.8-3	tidyselect_1.2.1
[91]	Biostings_2.74.1	knitr_1.49	IRanges_2.40.1
[94]	stats4_4.4.2	xfun_0.53	expm_1.0-0
[97]	Biobase_2.66.0	stringi_1.8.7	UCSC.utils_1.2.0
[100]	geodist_0.1.1	yaml_2.3.10	evaluate_1.0.3
[103]	codetools_0.2-20	BiocManager_1.30.26	cli_3.6.5
[106]	systemfonts_1.3.1	Rcpp_1.0.14	GenomeInfoDb_1.42.3
[109]	coda_0.19-4.1	parallel_4.4.2	phangorn_2.12.1
[112]	viridisLite_0.4.2	scales_1.4.0	crayon_1.5.3
[115]	combinat_0.0-8	rlang_1.1.6	cowplot_1.2.0
[118]	fastmatch_1.1-6	mnormt_2.1.1	